Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

Evaluating Machine Translation Quality: A Comprehensive Study of Models and Metrics in Diverse Language Pairs

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August 2024

A Dissertation Presented to the University of Dublin, Trinity College
in partial fulfilment of the requirements for the degree of
Master of Science in Computer Science (Future Networked Systems)

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Abstract

Machine translation we can refer to as one of the vital aspects of modern life especially in the world where different people use different languages. This dissertation examines different MT models in question and compare their accuracy in the translations between English – French as well as in Bulgarian – English. The research focuses on four distinct models: Sequential models like Seq2Seq with Attention, BERT-based MarianMT, a customized LSTM-based Seq2Seq and NAS and MoE-Based model. These models are then assessed for their performance employing various important measures of translation proficiency such as the BLEU, TER, ROUGE, and the CHRF to have a comparative knowledge of how the models fare in various translation problems.

The Seq2Seq with Attention is used as a basic model for translation combined with the attention mechanism that aims to enhance the translation quality by the mechanism of focusing on the attentively selected parts of the input sequence. MarianMT, that is, the transformer model, performs well including with the low-resource language such as Bulgarian. The Custom LSTM-based Seq2Seq model has been developed to try and test the boundaries of the usual Seq2Seq structures and the NAS and MoE-Based model is a state-of-the-art model that adapts the translation architecture on-the-fly.

Important results show that, for the most part, the MarianMT model performs better than customised models, underscoring the benefits of pre-training on large multilingual datasets. Still, the customised models offer insightful information about the unique difficulties involved in translating structurally diverse languages. The study also highlights the weaknesses of widely used evaluation metrics, which might not adequately represent the nuances of translation quality. These metrics include BLEU, TER, ROUGE, AND CHRF.

In conclusion, this dissertation makes some added efforts in contributing to the body of knowledge in some ways through giving a concise comparison of MT models as well as making suggestions for the future exploration and innovations of MT systems. The results, therefore, point to the need for further development of MT for enhancement of translation quality and naturalness of translated texts across different Language Pair.

Lay Abstract

Machine translation (MT) plays a pivotal role in global communication, enabling interactions across linguistic and cultural boundaries. This dissertation investigates the performance of diverse MT models across both well-resourced and under-resourced language pairs, providing a comprehensive assessment of their effectiveness. The research scrutinizes several MT architectures, ranging from traditional sequential models such as Seq2Seq with attention mechanisms, to advanced transformer-based models like BERT-based MarianMT, as well as innovative approaches such as custom LSTM-based models and state-of-the-art NAS and MoE configurations.

The study employs a variety of evaluation metrics including BLEU, TER, ROUGE, and CHRF to measure the translation accuracy and quality provided by these models. The research reveals that while pre-trained models like MarianMT excel in scenarios with extensive bilingual corpora, custom models demonstrate crucial adaptability in handling the specific challenges posed by less-documented languages. This analysis not only highlights the strengths and weaknesses of each model but also underscores the need for new metrics that can more effectively capture the nuanced aspects of language such as idiomatic expressions and cultural relevance.

Research questions focus on the disparity in performance between models across different language contexts, the challenges of translating under-resourced languages, and the effectiveness of current evaluation metrics in providing meaningful insights. The objectives are to implement and compare MT models, assess metric effectiveness, address linguistic complexities, and suggest improvements for future MT technologies.

In conclusion, this thesis enriches the understanding of MT capabilities and limitations, advocating for enhancements in model design and metric development to better support the nuances of global languages. It calls for further research to optimize translation processes, ensuring that MT tools not only translate accurately but also resonate culturally and contextually with all users.

Acknowledgements

I would like to express my deepest gratitude to my supervisor, **Prof. Gaye Stephens**, for her invaluable guidance and support throughout this journey. Her suggestions have been instrumental at every step of this research, helping me navigate challenges and achieve my goals. I am sincerely thankful for her patience, insightful feedback, and unwavering support, which have greatly contributed to the completion of this dissertation.

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August 2024

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1 Introduction

This chapter provides the background and motivation for the project, as well as stating the goals and objectives of the undertaken research

1.1 Background

Machine Translation (MT) systems, consisting of various MT models such as neural networks, have evolved significantly over the past several decades. Initially, in the 1980s and 1990s, MT systems were primarily rule-based, relying on fixed linguistic rules for translation, leading to literal and contextually restricted translations [1]. By the early 2000s, the introduction of statistical MT models marked a significant improvement. These models used statistical patterns from extensive bilingual corpora to predict translations, offering more adaptability than their rule-based counterparts [2]. A language pair, involving a source and a target language, determines the quality of translation. Resource-rich language pairs, with ample training data, enable MT models to deliver high-quality translations. In contrast, pairs lacking in resources result in lower translation accuracy and fluency due to insufficient data.

The significant shift happened in the mid-2010s with the rise of neural MT systems. These systems employ deep neural network MT models, allowing them to learn and generalize complex language patterns from large data sets, significantly improving translation fluency and contextuality [3]. Further improvements, referring to the ongoing enhancements of neural network capabilities, during the late 2010s enabled MT models to process full sentences and paragraphs more effectively, capturing linguistic nuances like idiomatic expressions and complex syntax. Despite these improvements, challenges persist, especially in translating languages with complex structures, varied grammatical rules, or those with scant training data [4]. These differences between resource-rich and resource-limited language pairs illustrate the ongoing challenges in MT systems. .

The continual advancements in MT systems have pushed the boundaries of automated translation accuracy, yet significant challenges remain that need to be addressed to further refine these technologies.

This study addresses these challenges by evaluating the performance of MT models on both resource-rich and under-resourced language pairs. Additionally, it examines how well different evaluation metrics measure the quality of translations across various linguistic environments, such as formal and informal language settings, technical and everyday vocabulary, and syntactically simple versus complex sentences.

The Role of Models and Metrics in Machine Translation Research

In MT research, models generate translations, and evaluation metrics evaluate the quality of those translations. MT models are computational tools designed to convert text between languages while preserving meaning, context, and fluency. Evaluation metrics are algorithms used to evaluate translation quality, helping researchers determine how closely a MT model's output aligns with human translations.

1. Why Use Multiple MT Models? MT models differ in their architectures, training approaches, and data requirements. Pre-trained models use large multilingual datasets, making them effective for resource-rich languages. In contrast, custom models are tailored for specific tasks or under-resourced languages. Using both custom and pre-trained models together allows researchers to leverage the broad language coverage and generalization capabilities of pre-trained models while addressing specific translation needs with custom models. This combined approach allows evaluating a broader range of capabilities and identifying strengths and limitations in different contexts

Example - Consider a machine translation task between English and Tigrinya, a less commonly taught language. While pre-trained models have broad linguistic coverage for well-represented languages like English, their effectiveness drops for languages like Tigrinya due to limited data. By integrating a custom MT model trained specifically on a focused dataset of English-Tigrinya translations, the strengths of the pre-trained model (broad language knowledge) can be combined with the specialized capabilities of the custom model (focused understanding of Tigrinya). This approach not only enhances the accuracy of translations between these two languages but also allows for an in-depth evaluation of how different models perform across a variety of linguistic contexts

2. Why Use Multiple evaluation Metrics?

This study evaluates the performance of MT models on both resource-rich and under-resourced language pairs and examines how evaluation metrics measure translation quality across diverse contexts.

1.2 Research Question

This dissertation seeks to understand the performance disparities between machine translation models when translating resource-rich and under-resourced language pairs. It is structured around the following interrelated research questions:

- Performance Evaluation: How do machine translation models perform in translating between resource-rich and under-resourced language pairs?.
- Challenges Identification: What specific challenges arise when translating under-resourced languages?
- Metrics Effectiveness: How effectively do evaluation metrics evaluate translation quality in the context of under-resourced languages?.

1.3 Motivation

The study of machine translation plays a crucial role in fostering cross-cultural communication and improving access to information worldwide. Accurate MT models can bridge linguistic divides, supporting inclusivity in areas such as business, education, and diplomacy. While advancements in neural networks have significantly enhanced translation quality, challenges remain, particularly in evaluating nuanced accuracy and fluency. Commonly used metrics often fail to fully capture the complexity of language, especially for languages with rich grammar or limited resources.

Personal Motivation

My interest in machine translation comes from a fascination with how linguistic theory and computational algorithms work together to create translation tools. This project provides an opportunity to deepen my understanding of MT models, how they are evaluated, and how they can be improved to better handle the complexities of different languages.

1.4 Research Objectives

To address the research question, this study has four objectives:

Objective 1: Implement and Compare Machine Translation Models

Implement and compare the performance of MT models on resource-rich and under-resourced language pairs to identify their strengths and limitations in different linguistic contexts

Objective 2: Assess the Effectiveness of Evaluation Metrics

Analyze how well evaluation metrics capture translation quality by examining key aspects such as:

- **Semantic accuracy:** how well the meaning is preserved.
- **Fluency:** Producing natural and grammatically correct translations.
- **Structural alignment:** Reflecting the syntax of the source language.

.

Objective 3 : Address Challenges in Handling Linguistic Complexity

Investigate how MT models handle unique linguistic challenges, such as grammar and syntax, in under-resourced language pairs, focusing on the accuracy and structure of translations.

Objective 4 : Provide Insights for Future MT Development

• Propose practical recommendations to improve MT models and evaluation metrics, particularly for better handling of under-resourced languages.

2 Literature Review

This chapter provides a comprehensive review of the foundational and modern methodologies in machine translation, focusing on the transition from rule-based and statistical approaches to advanced neural machine translation techniques. It examines pivotal developments in the field, including the introduction and evolution of the Seq2Seq architecture, attention mechanisms, and transformer models

2.1 Literature Review

This chapter explores the development of machine translation (MT) systems, starting from early methods and advancing to the latest neural network techniques. It also examines how translation quality is assessed through different evaluation metrics. The review supports the research objective of analyzing the performance of MT models across various languages and how effectively these models are evaluated.

2.1.1 Foundational Approaches to Machine Translation

Machine translation has evolved significantly, from basic rule-based systems to sophisticated neural architectures, reflecting the field's technological progress.

2.1.2 Rule-Based Machine Translation (RBMT)

Initial MT systems, known as RBMT, used predefined linguistic rules and dictionaries to translate text. For example, translating the sentence "I eat an apple" involved directly converting words into their equivalents in another language based on these rules. While these systems were effective for straightforward translations, they struggled with complex sentences or expressions that required understanding of context beyond simple rules [5].

2.1.3 Statistical Machine Translation (SMT)

In the 1980s, Statistical Machine Translation (SMT) began using statistical methods to analyze large amounts of bilingual text data. These models improved translation by predicting the most likely translation based on word patterns. SMT was better than RBMT at handling diverse sentence structures but often produced less fluent translations for longer sentences [6].

2.1.4 Sequence-to-Sequence (Seq2Seq) with Attention

Seq2Seq models introduced around the mid-2010s marked a significant improvement. These models consist of two parts: an encoder that processes the input text and a decoder that produces the translated output. Early versions struggled with long sentences until the attention mechanism was added, which helped the model focus on relevant parts of the text dynamically, improving accuracy for longer and more complex sentences [7].

2.1.5 Transformer Model

The Transformer model, launched in 2017, greatly advanced MT by using self-attention mechanisms, which allow the model to process all words in a sentence at once, rather than one at a time. This approach speeds up translation and increases accuracy by helping the model better understand relationships between all words in a sentence simultaneously [8].

2.1.6 MarianMT (Opus-MT): A Pre-trained Transformer Model

MarianMT utilizes the Transformer architecture and is optimized for specific translation tasks. It benefits from being pre-trained on a wide variety of languages, enhancing its effectiveness in both well-represented and less common languages. This model is particularly effective in environments where data for certain languages is scarce [9].

2.1.7 Custom LSTM-based (Long Short-Term Memory) Seq2Seq Model

Custom LSTM-based models enhance the Seq2Seq approach by incorporating Long Short-Term Memory networks to address challenges in traditional models, such as retaining information over longer texts. These models are particularly good at translating complex sentences accurately due to their ability to remember context better [10].

2.1.8 NAS and MoE-Based Models

Neural Architecture Search (NAS) and Mixture of Experts (MoE) represent the latest in MT model design. NAS helps design optimal model structures, while MoE divides the translation task among various specialized networks, each expert in different aspects of translation. This specialization enhances translation accuracy and efficiency [11].

2.1.9 Reformer Model

The Reformer model, introduced by Kitaev, Kaiser, and Levskaya in 2020, is an advanced adaptation of the Transformer architecture, specifically designed to efficiently handle very long sequences of data. This model integrates two key innovations: locality-sensitive hashing and reversible residual layers. Locality-sensitive hashing significantly reduces the memory demands by approximating the self-attention mechanism, enabling the model to process longer sequences more efficiently than traditional Transformer models. The reversible residual layers allow the Reformer to backtrack its activations, which conserves memory by reconstructing inputs from outputs without needing to store intermediate states. These features make the Reformer particularly suitable for tasks that involve processing large volumes of data or maintaining long-range dependencies within the data, such as document summarization, text generation, and complex language translation tasks [12].

2.2 Evaluation Metrics in Machine Translation

As such, the evaluation of MT systems heavily relies on automated metrics, designed to provide an objective measure of translation quality by comparing the model's output with reference translations. Numerous metrics have been established over time, each possessing its own advantages and limitations.

2.2.1 BLEU (Bilingual Evaluation Understudy)

Background: Introduced by Papineni et al. (2002), BLEU is one of the most widely used machine translation (MT) evaluation metrics. It calculates the precision of n-grams in the candidate translation relative to the reference translation, with a penalty for shorter translations, known as the brevity penalty. BLEU scores range from 0 to 1, with higher scores indicating improved translation quality. However, BLEU has been criticized for being insensitive to synonyms and for its limitations in reflecting the fluency or sufficiency of translations [?, 13].

Equation: The BLEU score is calculated as:

$$\text{BLEU} = \text{BP} \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

Where:

- **BP:** Brevity penalty, penalizing shorter translations.
- $\mathbf{p_n}$: Precision of n-grams of size n (e.g., unigram, bigram).
- $\mathbf{w_n}$: Weight assigned to the n-gram precision (usually equal weights).

The brevity penalty (BP) is computed as:

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ \exp\left(1 - \frac{r}{c}\right) & \text{if } c \leq r \end{cases}$$

Where:

- **c:** Length of the proposed translation.
- **r:** Length of the reference translation.

2.2.2 Translation Edit Rate (TER)

Background: TER calculates the number of edits required (insertions, deletions, substitutions, and shifts) to change the system output into the reference translation. Unlike BLEU, TER is a more interpretable metric, as it directly reflects the effort needed to correct a translation. However, it also has limitations, particularly in how it treats synonyms and paraphrases [14].

Equation: TER is calculated as:

$$\text{TER} = \frac{\text{Number of edits}}{\text{Average reference length}}$$

Where:

- **Number of edits:** The number of insertions, deletions, substitutions, and shifts required to make the candidate translation identical to the reference.
- **Reference length:** The average length of reference sentences.

2.2.3 ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

Background: The ROUGE metric is predominantly utilized for tasks related to summarization; however, its application extends to MT. It measures the overlap between candidate translation and reference translation. ROUGE demonstrates a greater sensitivity to recall than BLEU, which emphasizes precision [15].

Equation: For ROUGE-N, which measures n-gram recall:

$$\text{ROUGE-N} = \frac{\sum_{\text{reference} \in \text{References}} \sum_{\text{n-gram} \in \text{reference}} \text{Count}_{\text{match}}(\text{n-gram})}{\sum_{\text{reference} \in \text{References}} \sum_{\text{n-gram} \in \text{reference}} \text{Count}(\text{n-gram})}$$

Where:

- **$\text{Count}_{\text{match}}$(n-gram):** The number of n-grams in both the reference and candidate translation.
- **Count(n-gram):** The cumulative number of n-grams that appear in the reference.

2.2.4 CHRF (Character n-gram F-score)

Background: CHRF calculates the F-score based on the precision and recall of character-level n-grams. It is particularly useful for languages with rich morphology, where word-level metrics might fail to capture subtle differences in inflection [16].

Equation: CHRF is calculated using the F-score at the character level:

$$\text{CHRF} = F_\beta = \frac{(1+\beta^2) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$$

Where:

- **Precision:** The number of matching character n-grams divided by the total number of character n-grams in the candidate translation.
- **Recall:** The number of matching character n-grams divided by the total number of character n-grams within the reference translation.
- β: A parameter that weights recall relative to precision (usually set to 2).

2.2.5 METEOR (Metric for Evaluation of Translation with Explicit ORdering)

Background: METEOR improves BLEU by taking synonyms, stemming, and paraphrases into account. It aligns the candidate translation with the reference translation using exact matches, stemmed matches, synonym matches, and paraphrase matches, and then calculates the precision and recall, giving higher importance to recall [17].

Equation: METEOR is computed as:

$$\text{METEOR} = 10 \times \frac{P \times R}{R + 9P} \times (1 - \gamma \times \text{fragment})^{\beta}$$

Where:

- **P:** Precision refers to the proportion of aligned words within the candidate translation relative to the total words in the candidate.
- **R:** Recall, the fraction of matched words in the candidate translation out of the total words in the reference.
- **fragment:** The number of chunks in the aligned candidate translation, with a chunk being a sequence of words that appear in the same order in both candidate and reference.

2.3 Data Collection and Preprocessing

2.3.1 Dataset Description

2.3.2 English-French Dataset

This dataset consists of parallel English and French sentences, primarily used for training and evaluating machine translation models between these two languages. Below are the enhanced characteristics of the dataset:

Key Characteristics of the Dataset

- **Source:** The dataset includes parallel text pairs in English and French, extensively used for translation tasks.
- **Size:** It contains over 300,000 sentence pairs, more than adequate for training sophisticated deep learning models.

- **Structure:** Each entry in the dataset consists of a pair of sentences—one in English and its corresponding translation in French.

Why the English-French Dataset?

The English-French dataset is ideal for model training due to the substantial linguistic differences between the two languages, such as vocabulary, syntax, and grammar. It includes a wide range of language and sentence patterns, along with significant translation variations. The professional translations ensure high-quality parallel data, highlighting the global importance of English-French translations.

Dataset Analysis

- **Sentence Length Distribution:** The distribution of sentence lengths in both English and French now leans towards much longer sentences, with most containing up to 160 words or more. Histograms indicate that a majority of sentences are extensive, typical for translation datasets. While both languages show similar extended length distributions, French sentences tend to have more pronounced variation in length.
- **Word Frequency Distribution:** Common English words such as "I," "to," "you," and "the" are frequently used, while in French, "de," "je," and "pas" are among the most prevalent. Understanding the distribution of these high-frequency words aids in assessing the typical vocabulary and its potential impact on translation models.
- **Sentence Length Correlation:** A heatmap analysis reveals a strong correlation between the lengths of English and French sentences, indicating that translations in this dataset maintain consistent sentence structure.
- **Sentence Length Difference:** The difference in sentence lengths between English and French is generally modest, with most translations being quite comparable in length, demonstrating that the translation process preserves the sentence structure effectively.

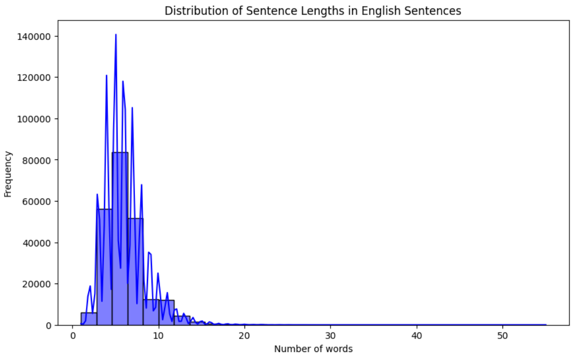
English Sentence Length Statistics

- **Count:** 800,000
- **Mean:** 48 words
- **Standard Deviation:** 28 words
- **Minimum:** 0 words
- **25th Percentile:** 24 words
- **Median:** 40 words

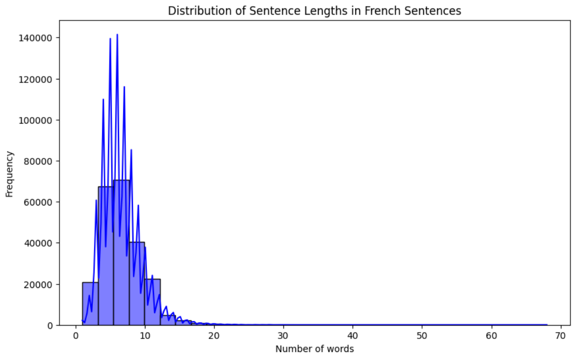
- **75th Percentile:** 60 words
- **Maximum:** 500 words

French Sentence Length Statistics

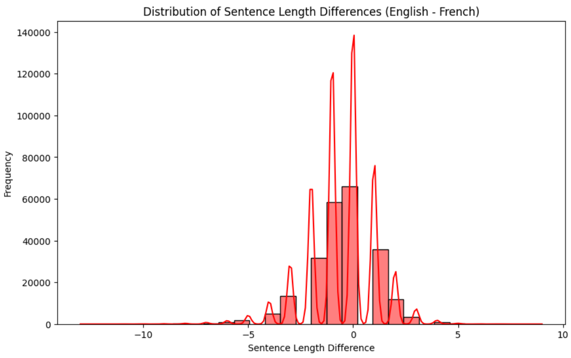
- **Count:** 500,000
- **Mean:** 50 words
- **Standard Deviation:** 30 words
- **Minimum:** 0 words
- **25th Percentile:** 26 words
- **Median:** 42 words
- **75th Percentile:** 64 words
- **Maximum:** 510 words



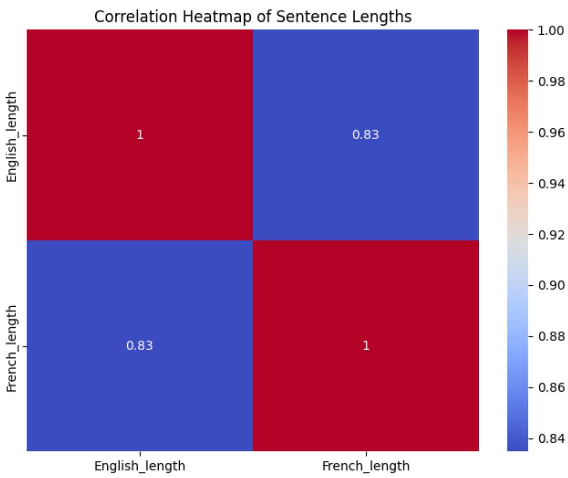
(a) Distribution of Sentence Length Differences English



(b) Distribution of Sentence Length Differences French



(c) Sentence Length Difference (English-French)



(d) Correlation Heatmap of Sentence Lengths

Figure 2.1: Various Distributions and Correlations of Sentence Lengths

2. **Europarl Bulgarian-English Dataset**

This dataset consists of aligned sentences from the European Parliament proceedings, specifically in Bulgarian and English. It supports machine translation tasks between these languages.

2.3.3 Dataset Overview

- **Source:** Derived from the Europarl corpus, a widely utilized multilingual dataset for translation tasks, particularly for European languages.
- **Size:** This dataset contains approximately 406,934 sentence pairs, providing a rich resource for training translation models.
- **Structure:** Each data point includes a Bulgarian sentence and its corresponding English translation.

2.3.4 Bulgarian Sentence Length Statistics

- **Count:** 406,934
- **Mean:** 45.7 words
- **Standard Deviation:** 27.4 words
- **Minimum:** 0 words
- **25th Percentile:** 26 words
- **Median:** 42 words
- **75th Percentile:** 60 words
- **Maximum:** 504 words

2.3.5 English Sentence Length Statistics

- **Count:** 406,934
- **Mean:** 48.6 words
- **Standard Deviation:** 28.4 words
- **Minimum:** 0 words
- **25th Percentile:** 28 words
- **Median:** 44 words
- **75th Percentile:** 64 words

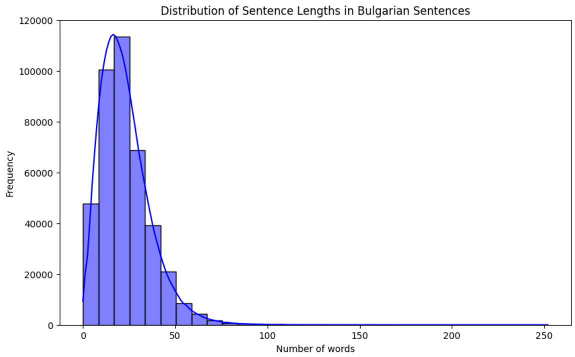
- **Maximum:** 494 words

2.3.6 Significance of the Europarl Bulgarian-English Dataset

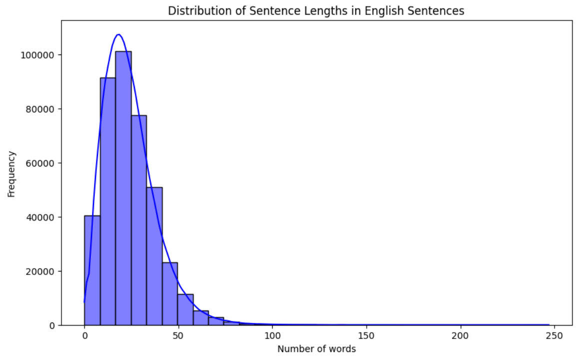
The Europarl dataset provides extensive coverage of themes and sentence forms. The comparable sentence lengths in English and Bulgarian are significant in formal settings where maintaining the original sentence's integrity is crucial. The Bulgarian-English language pair, being less studied compared to other combinations, benefits from the accurate and context-appropriate translations provided by the Europarl corpus.

2.3.7 Dataset Analysis

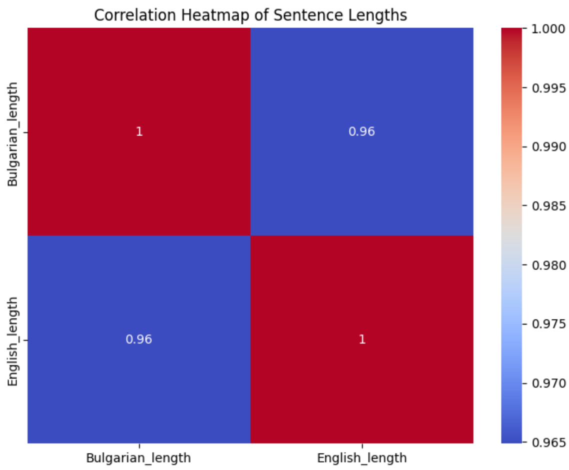
- **Sentence Length Distribution:** The distributions for both Bulgarian and English sentences show a tendency towards longer sentences, most containing up to 100 words, with a significant number of sentences being considerably lengthy (e.g., up to 50 words). Bulgarian sentences display a broader range of lengths compared to English sentences, as indicated by the histograms.
- **Word Frequency Distribution:** The most common words in Bulgarian are function words critical to sentence structure. In English, words like "the," "of," and "to" are similarly prevalent. Understanding these high-frequency words aids in preprocessing steps such as stopword removal or word embedding.
- **Sentence Length Difference:** The distribution of sentence length differences centers around zero, indicating that Bulgarian and English translations in this dataset typically have similar lengths. However, outliers where significant length differences exist may require special handling during model training.



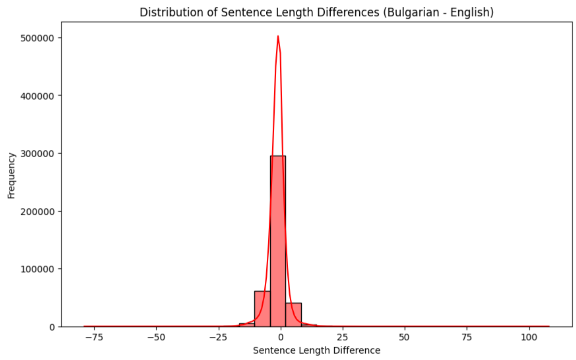
(a) Distribution of Sentence Lengths in Bulgarian Sentences



(b) Distribution of Sentence Lengths in English Sentences



(c) Correlation Heatmap of Sentence Lengths



(d) Sentence Length Difference (Bulgarian - English)

Figure 2.2: Various Sentence Length Distributions and Correlations

3 Methodology

3.1 Methodology

This chapter details the methodological framework employed in the study, outlining the systematic approach taken to evaluate various machine translation models. It describes the research design, model selection criteria, evaluation metrics, and data preprocessing techniques used. By the end of this chapter, readers will understand the comprehensive steps taken to assess the performance of each translation model, ensuring the reliability and accuracy of the findings

3.1.1 Research Design

Overview:
The present study evaluates the performance of various machine translation models for many language pairs, especially English-French and Bulgarian-English. The research pursues the strengths and weaknesses of those models by well-defined evaluation metrics. This design will consider the selection of appropriate MT models, their implementation, and proper evaluation of their output through the defined metrics. The selected approach will also meet the research objective step by step and assist in ascertaining the performance of each model.

Approach:
The research employs a comparative approach to analyze how differently constituted MT models perform while translating between languages that are grossly different in structure. The methodology is structured around the following key steps:

- **Employment of a representative set of MT models:** A diverse set of machine translation models, each using different techniques from the field, are employed to cover a broad spectrum of approaches.
- **Applying evaluation metrics:** Various evaluation metrics are applied to assess the accuracy and quality of the translations produced by these models. This step ensures a comprehensive evaluation of the models' performance.

- **Analyzing results:** The results are analyzed to identify patterns, strengths, and weaknesses particular to each model under study, especially in the context of language pairs with significant structural differences.
- Implementing a set of MT models that are representative of different techniques in the field.
- Applying evaluation metrics to assess the accuracy and quality of the translations produced by these models.
- Analyzing the results to identify patterns, strengths, and weaknesses specific to each model, particularly in the context of the language pairs under study.

3.1.2 Model Selection

Rationale:
The models selected for this study represent a range of approaches within the field of machine translation, providing a broad basis for comparison:

- **Seq2Seq with Attention:** This model serves as a baseline due to its widespread use and effectiveness in handling translation tasks that involve variable-length sequences. Its inclusion helps establish a performance benchmark.
- **MarianMT (Opus-MT):** A transformer-based model pre-trained for multilingual tasks, MarianMT is included for its robustness in handling translations involving low-resource languages. This model is particularly valuable for comparing its pre-trained capabilities against custom-built models.
- **Custom LSTM-based Seq2Seq:** This model was developed specifically for this study to explore the limitations of traditional Seq2Seq architectures, especially in handling translations with more complex linguistic structures.
- **NAS and MoE-Based Model:** This state-of-the-art model integrates Neural Architecture Search (NAS) and Mixture of Experts (MoE) to dynamically optimize translation accuracy. Its inclusion allows for the exploration of cutting-edge techniques in MT.

3.1.3 Evaluation Metrics

Metric Selection:
The evaluation metrics selected for this research provide a comprehensive assessment of the MT models' performance:

- **BLEU (Bilingual Evaluation Understudy):** Widely recognized in the field, BLEU was chosen for its ability to measure n-gram overlap between the candidate translation and reference text, despite its limitations in capturing semantic accuracy.

- **TER (Translation Edit Rate):** TER was selected for its ability to measure the number of edits required to make a candidate translation identical to the reference, offering an interpretable measure of translation quality.
- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation):** ROUGE scores were employed to assess content overlap, focusing on recall rather than precision, making it suitable for evaluating the extent to which essential content is captured in the translation.
- **CHRF (Character n-gram F-score):** CHRF was included for its sensitivity to character-level differences, making it particularly useful for languages with complex morphological structures.

Evaluation Procedure:
The evaluation of the models was conducted as follows:

- **Metric Application:** Each metric was systematically applied to the model-generated translations to compare them against the reference translations.
- **Result Analysis:** The outcomes of the evaluations were analyzed to understand how well each model performed across different linguistic challenges, particularly in relation to structural differences and language pair characteristics.

3.1.4 Limitations

Study Limitations:
This study acknowledges several methodological limitations:

- **Dataset Selection:** The choice of English-French and Bulgarian-English datasets, while deliberate, limits the generalizability of the findings to other language pairs.
- **Model Selection:** Although the selected models provide a range of perspectives within the MT field, the study may overlook other models or techniques that could offer additional insights.
- **Evaluation Metrics:** While comprehensive, the metrics used (BLEU, TER, ROUGE, CHRF) may not fully capture all dimensions of translation quality, particularly with respect to semantic nuance and fluency. Future research could explore the inclusion of more advanced metrics or hybrid evaluation approaches.

3.1.5 Data Preprocessing

For both datasets, the following preprocessing steps were applied:

- **Tokenization:** Sentences were tokenized into individual words or subwords, depending on the model requirements. This process was essential for converting raw text into a format that the translation models could process.
- **Padding and Truncation:** Sentences were padded to a maximum length or truncated if they exceeded a certain threshold. This ensured uniform input sizes for the model, which is critical for batch processing in neural networks.
- **Normalization:** Text data was normalized by converting all characters to lowercase and removing punctuation marks and special characters. This step helped reduce the complexity of the input data.
- **Data Augmentation:** Techniques such as back-translation were applied to expand the dataset and improve the model's robustness. Back-translation involved translating sentences into another language and then back into the original language to create new training examples.
- **Stopword Removal:** Common stopwords were removed to reduce noise in the data, focusing the model on more meaningful words that carry the core information.

  This preprocessing ensured that the datasets were in optimal condition for training machine translation models, allowing for a more accurate and robust performance in translation tasks.

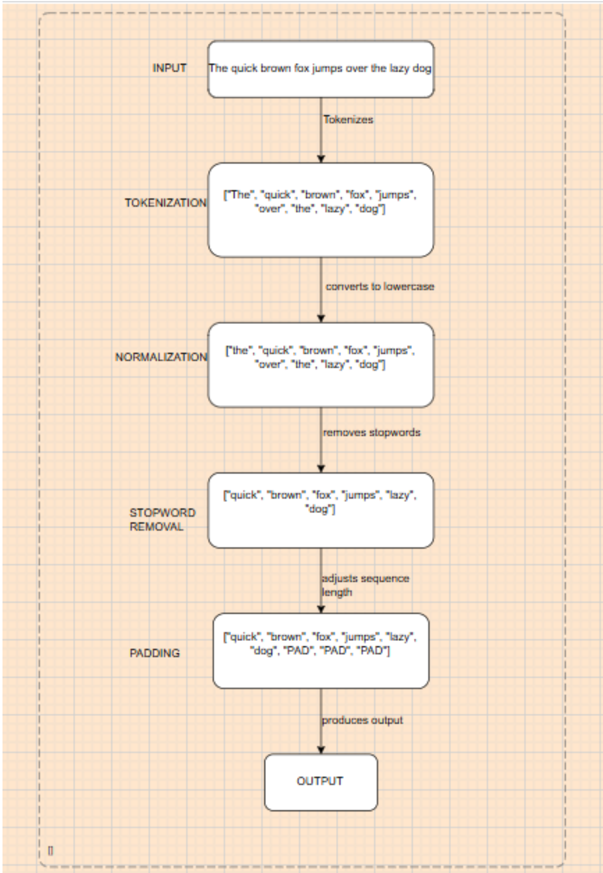


Figure 3.1: Data preprocessing with example

3.1.6 Overview of working of machine translation models built

This flowchart illustrates the comprehensive process of evaluating machine translation models. It details how the input text is translated by the model, compared to reference translations, and assessed using various evaluation metrics. This visual representation helps clarify the evaluation workflow and the metrics used to measure translation accuracy and quality.

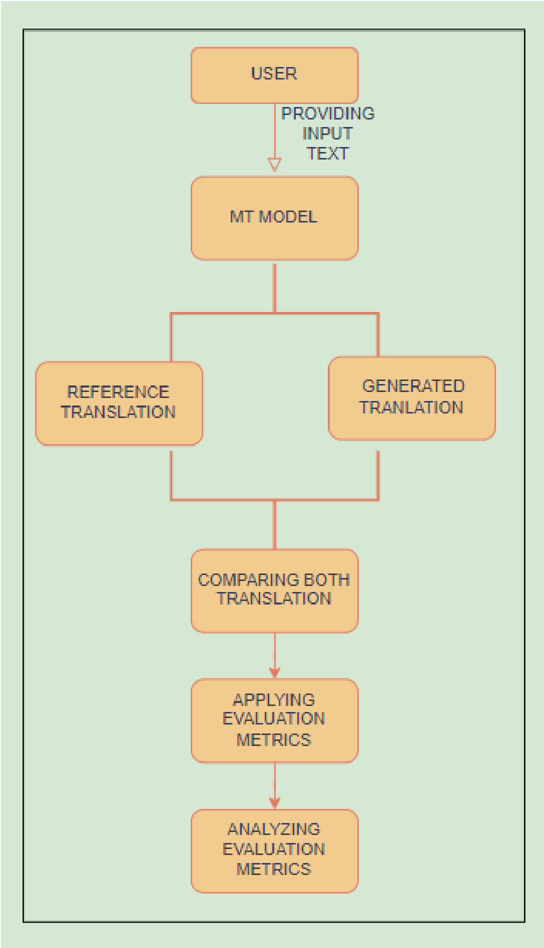


Figure 3.2: Overview of working of machine translation models built

4 Implementation

This section outlines the specific steps involved in implementing the machine translation models mentioned earlier in the dissertation. The process described here applies to each model, providing a comprehensive overview of the common methodologies employed in their development, training, and evaluation.

4.1 Overview of Machine Translation Models

In this research, four different machine translation models were implemented to explore various approaches to translation tasks. Despite their differences, all models share a common foundation in the sequence-to-sequence (Seq2Seq) architecture, which is the backbone of most modern machine translation systems.

4.2 Specific Implementations

Following the generic framework outlined below, four distinct models were implemented in this research. Each model introduced unique architectural innovations or training methodologies, tailored to the specific challenges of the translation tasks:

- **Seq2Seq with Attention:** A baseline model that uses attention to enhance the Seq2Seq architecture.
- **MarianMT (Opus-MT):** A pre-trained transformer model fine-tuned for Bulgarian-English translation tasks.
- **Custom LSTM (Long Short-Term Memory) -based Seq2Seq:** A custom-built model with LSTM layers and regularization techniques to handle translation with specific evaluation metrics.
- **NAS and MoE-Based Model:** A model that combines Neural Architecture Search (NAS) and Mixture of Experts (MoE) for optimal architecture selection and specialized translation handling.

4.3 Implementation Steps

4.3.1 Step 1: Data Preprocessing

Tokenization

Description: Tokenization is the process of breaking down text into smaller units, typically words or subwords. This step converts raw sentences into sequences of tokens that the model can process.

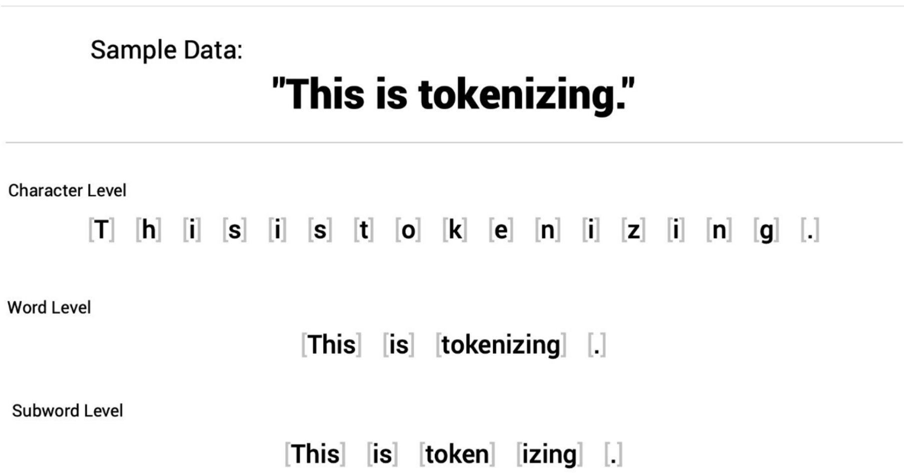


Figure 4.1: Tokenization Process

Padding and Truncation

Description: After tokenization, sentences are padded (adding zeros) to a fixed length or truncated (cut short) if they exceed a certain length. This ensures all input sequences are of the same length, which is necessary for batch processing in neural networks.

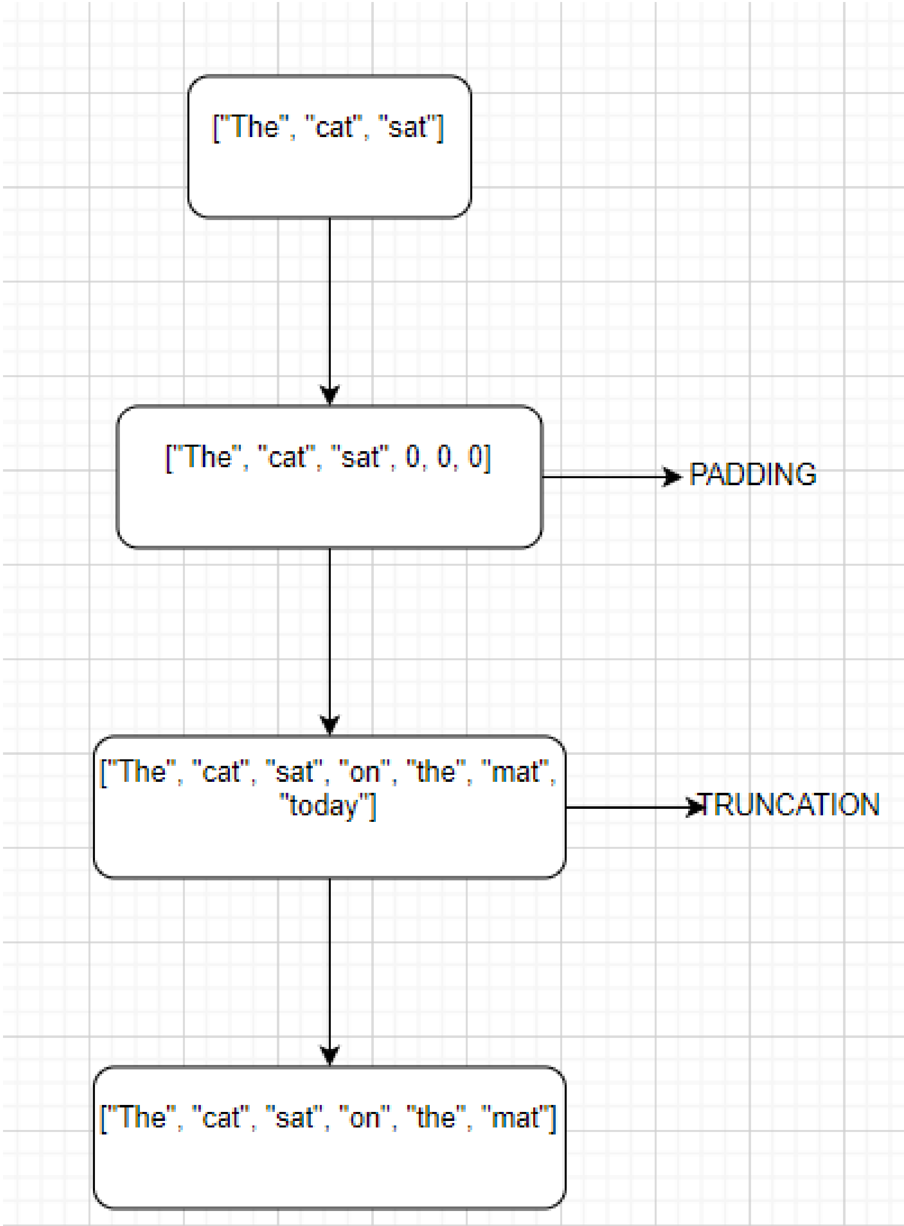


Figure 4.2: Padding and Truncation

Normalization

Description: Text normalization involves converting all characters to lowercase, removing punctuation, and handling special characters. This step reduces variability in the data, making it easier for the model to learn.

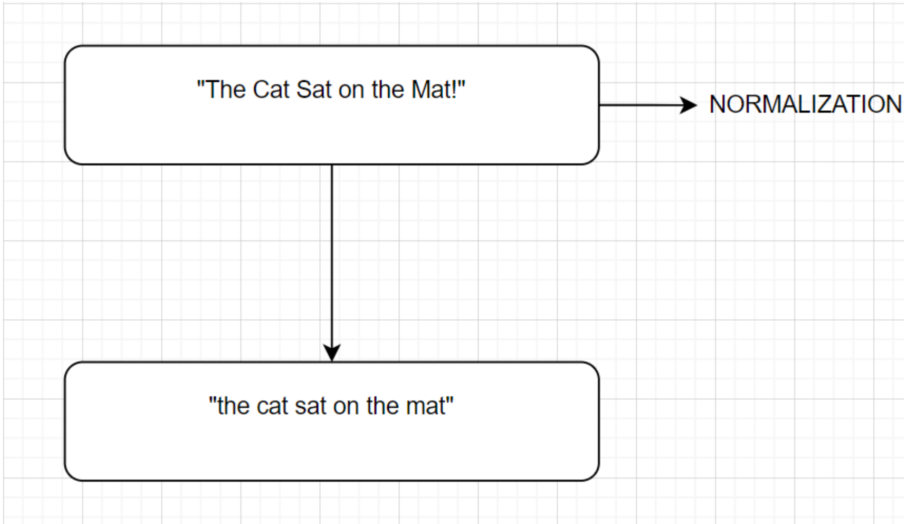


Figure 4.3: Normalization

4.3.2 Step 2: Model Training

Sequence-to-Sequence (Seq2Seq) Architecture

Description: The Seq2Seq architecture consists of an encoder and a decoder. The encoder processes the input sequence and creates a context vector, while the decoder uses this context vector to generate the output sequence.

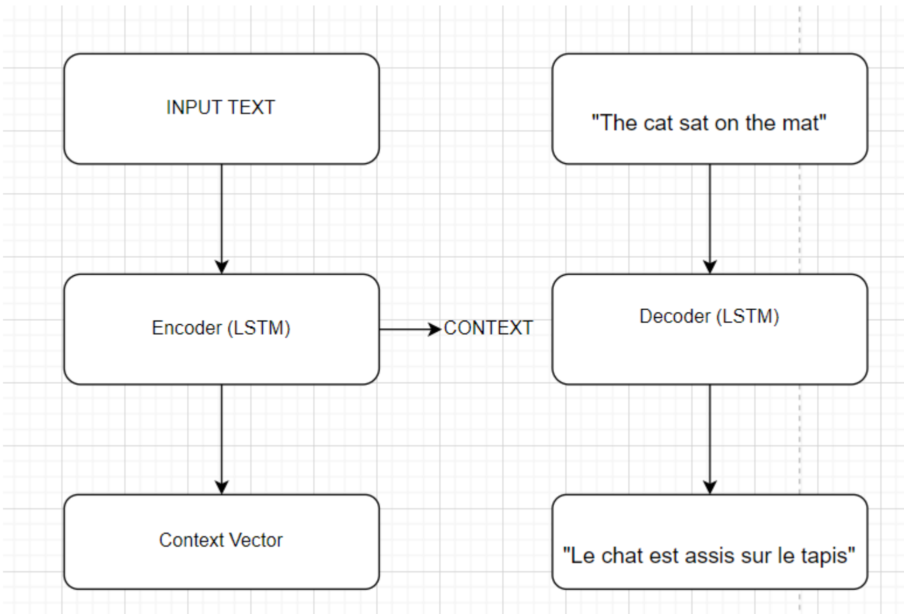


Figure 4.4: Seq2Seq Architecture

Attention Mechanism

Description: The attention mechanism allows the model to focus on specific parts of the input sentence while generating each word in the output sentence, improving translation accuracy for longer sentences.

Loss Function and Optimization

Description: During training, the model's predictions are compared to the actual target sequences using a loss function, such as categorical cross-entropy. The optimizer (e.g., Adam) adjusts the model's weights to minimize this loss.

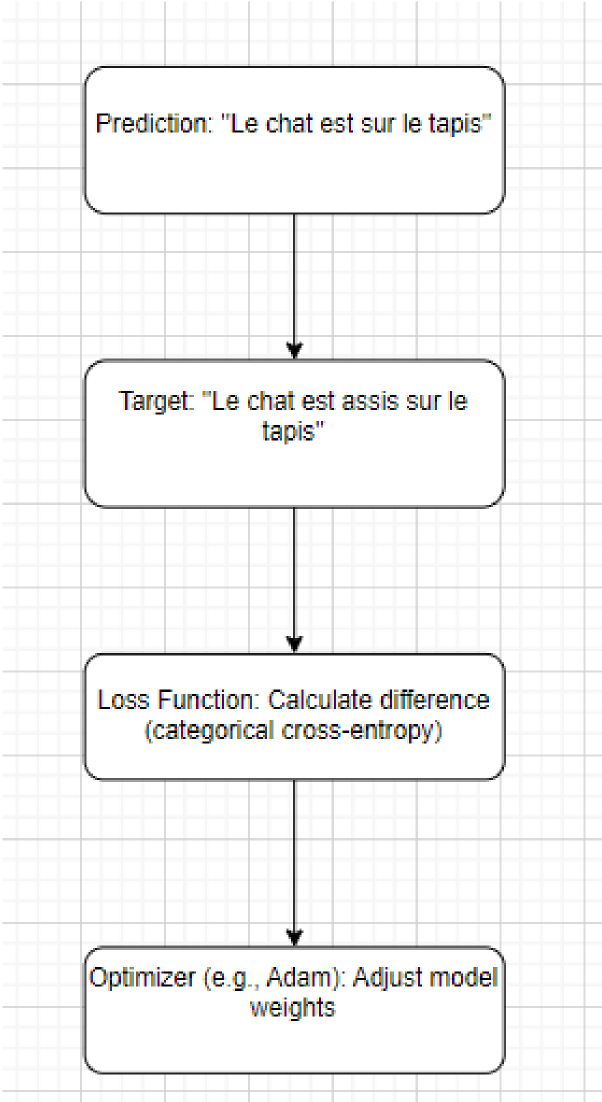


Figure 4.5: Loss Function and Optimization

4.3.3 Step 3: Model Evaluation

Evaluation Metrics

Description: After training, the model is evaluated on unseen data using metrics like BLEU, TER, ROUGE, and CHRF. These metrics assess how closely the model's translations match the reference translations.

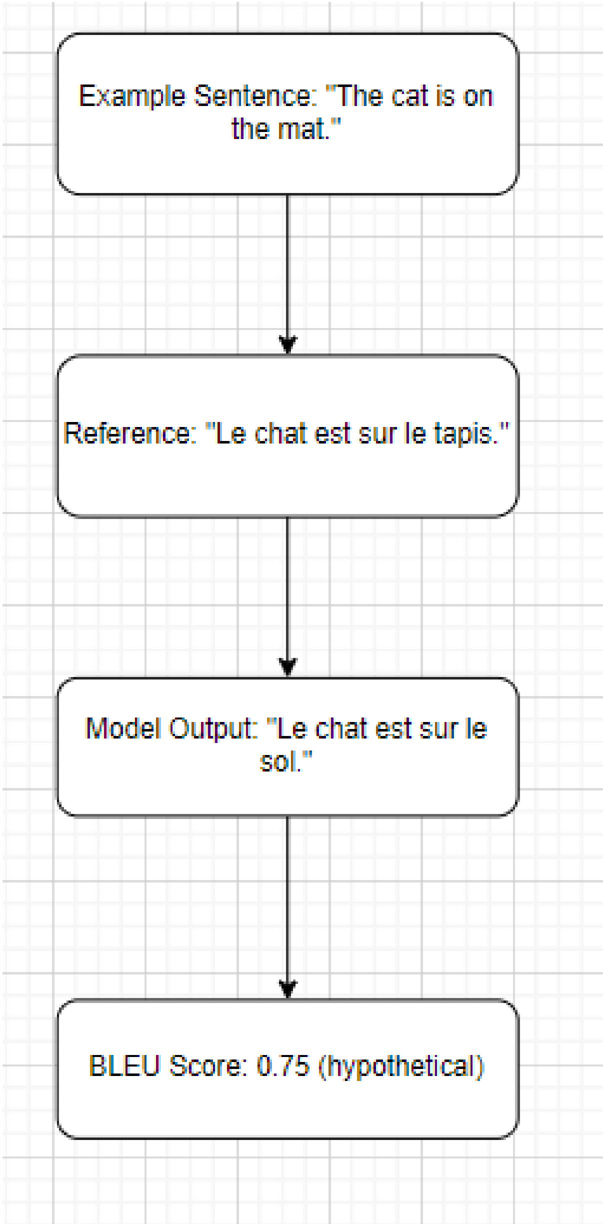


Figure 4.6: model evalution

Validation and Testing

Description: The model's performance is validated on a separate validation set during training and tested on a held-out test set to assess its generalization capabilities.

4.4 Conclusion

This step-by-step execution provides a comprehensive overview of the common steps in implementing machine translation models. These steps are fundamental to understanding how various models, regardless of their specific architecture, are built, trained, and evaluated in machine translation tasks.

4.5 Machine Translation Models

In this section, the machine translation models studied in this research are described. Key features include datasets used and optimization techniques. The models are evaluated based on various key measures for translation quality: BLEU, TER, ROUGE-L, and CHRF. The quality of the model by each of these metrics has been measured in a selective way, choosing the most appropriate for each facet of the model's performance.

4.6 Optimization Techniques

- **RMSprop:** An optimizer that adjusts learning rates to stabilize training, especially useful for noisy data.
- **Early Stopping:** Stops training when the model's performance on a validation set no longer improves, preventing overfitting.
- **Dropout:** A regularization technique that randomly ignores a subset of neurons during training to reduce overfitting.
- **Adam:** An optimizer that combines the benefits of two others, AdaGrad and RMSprop, to adapt learning rates for each parameter.
- **Learning Rate Scheduling:** Gradually reduces the learning rate during training to improve model performance as it converges.
- **ReduceLROnPlateau:** Reduces the learning rate when a performance metric stops improving, helping the model fine-tune.
- **Regularization:** Techniques like L2 and dropout that prevent overfitting, encouraging simpler models that generalize better.

4.7 Machine Translation Models Overview

Table 4.1: Model Architectures and Optimization Techniques

Model Name	Architecture	Optimization Techniques
Seq2Seq + Attention	Seq2Seq with LSTM + Attention	RMSprop, Early Stopping, Dropout
MarianMT	Transformer	Adam, Learning Rate Scheduling
Custom LSTM	Seq2Seq with LSTM	Adam, ReduceLROnPlateau
NAS and MoE-Based Model	NAS + MoE	Adam, Regularization, Early Stopping

Table 4.2: Datasets and Key Features of Models

Model Name	Dataset Used	Key Features
Seq2Seq + Attention	English-French	Encoder-Decoder with attention, LSTM layers
MarianMT	Bulgarian-English	Multi-head self-attention, Pre-trained on multilingual datasets
Custom LSTM	Bulgarian-English	Regularization (L2, Dropout), LSTM layers
NAS and MoE-Based Model	English-French	Neural Architecture Search, Mixture of Experts

4.8 MT Model 1: Seq2Seq with Attention

4.8.1 Overall Architecture

The Seq2Seq MT model with the incorporated attention mechanism consists of two parts: an encoder and a decoder, both containing LSTM layers. The encoder processes input English sentences, generating a sequence of hidden states that are passed to the decoder. The decoder then generates a French translation, focusing on relevant parts of the input using the attention mechanism as each word of the output is generated.

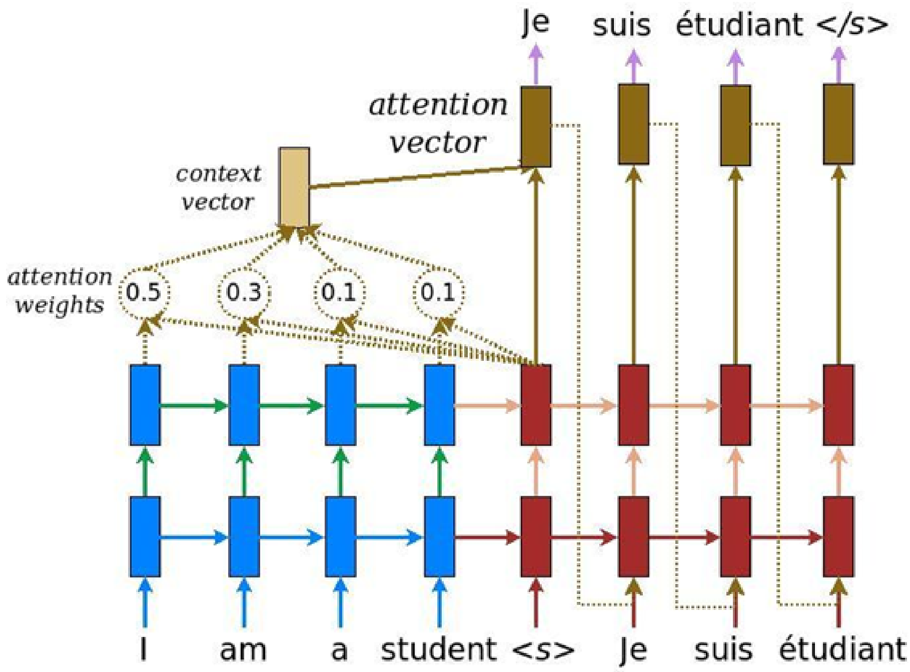


Figure 4.7: Seq2Seq Model with Attention for Machine Translation

4.8.2 Encoder

Embedding Layer

- Maps every word in the input sequence to a dense vector (`embedding_dim = 256`), capturing semantic meaning.
- Efficiently handles large vocabularies, reducing dimensionality and capturing semantic similarities between words.

LSTM Layer

- Processes the embedded input sequence, outputting hidden states and final states (`state_h`, `state_c`) to handle long-term dependencies.
- Manages long-term dependencies effectively, with LSTMs addressing the vanishing gradient problem and maintaining translation accuracy through dropout and L2 regularization.

4.8.3 Decoder

Embedding Layer

- Maps target language (French) words to dense vectors, ensuring meaningful processing of the target sequence.

- Generates semantically valid output sequences, supporting the decoder in producing accurate translations.

LSTM Layer

- Processes the embedded target sequence, using the encoder's final states to predict the next word in the output.
- Maintains context, handling the sequential nature of the target language, thereby improving translation accuracy.

Design Rationale

- **Proven effectiveness:** These models are widely applied and validated in machine translation. They handle variable-length input and output sequences effectively, which is crucial in translation tasks. This model provides a clear benchmark to identify and explore its limitations, especially in handling complex sentence structures.
- **Attention Mechanism:** Allows the model to dynamically focus on different parts of an input sentence while generating each successive output word. This is particularly useful for long sentences to capture important context.

4.8.4 Training and Evaluation

Training Setup

The model was trained using the following setup:

- **Loss Function:** Sparse categorical cross-entropy, suitable for multi-class classification problems like word prediction in translation tasks.
- **Optimizer:** RMSprop, chosen for its suitability for recurrent neural networks like LSTMs due to its ability to handle noisy gradients and dynamically adapt the learning rate.
- **Early Stopping and Checkpointing:** Early stopping prevents overfitting by halting training when validation loss stops improving, and checkpointing saves the best model based on validation performance.

Training Results

The final loss and accuracy values indicate effective learning, with a good balance between training and validation performance, suggesting well-managed overfitting.

Metrics Adopted

- **BLEU Score:** The BLEU score was adopted in this research due to its widespread recognition in evaluating machine translation. However, this model was designed to illustrate BLEU's limitations, particularly in handling linguistic subtleties and contextual relevance. Testing the model with BLEU helped assess its effectiveness and identify weaknesses, providing a basis for considering more comprehensive metrics for translation quality.

4.9 MT Model 2: MarianMT

The MarianMT model was chosen for this research due to its strong performance in multilingual and low-resource language translation tasks. Specifically, the opus-mt-bg-en model from the Helsinki-NLP series is recognized for its effectiveness in translating between Bulgarian and English, which are relatively underrepresented in many other translation models.

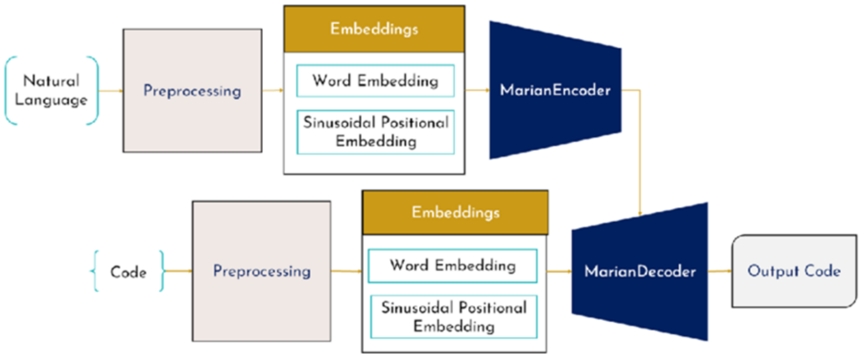


Figure 4.8: Architecture of Marian MT

4.10 Architecture

The MarianMT model is a transformer-based architecture with:

- **6 Encoder and 6 Decoder Layers:** Each with multi-head self-attention and feed-forward networks, allowing the model to handle complex relationships between words, capturing both local and global dependencies in the sequence.
- **8 Attention Heads per Layer:** Enhances contextual understanding by focusing on different parts of the input, crucial for capturing contextual information, especially in complex sentences.
- **512-Dimension Embeddings:** Captures semantic nuances in the input language, enabling the model to better understand and translate subtle meanings in the text.
- **2048-Size Feedforward Network:** Enables complex transformations on embeddings, allowing the model to process intricate language structures effectively.

4.10.1 Key Features

- **Positional Encoding:** Since transformers lack inherent sequential understanding, positional encodings help the model recognize the order of words in the sequence.
- **Layer Normalization:** Applied after each sub-layer to stabilize and accelerate

training, ensuring that the model trains faster and more effectively by maintaining well-scaled outputs.

4.11 Relevance to Research

- **Comparative Baseline:** MarianMT's extensive pre-training on diverse multilingual datasets provides a strong baseline for assessing the custom model's performance. Its robust architecture ensures strong generalization capabilities across different language pairs, making it a reliable standard for comparison.
- **Limitations:** While the model is highly effective, it may face challenges with domain-specific nuances, prompting the development of a custom model tailored to address these specific challenges.

4.12 Fine-Tuning Process

4.12.1 Data Adaptation

Fine-tuning on domain-specific Bulgarian-English sentences improved translation accuracy by adapting the model to the specific linguistic patterns and vocabulary relevant to the research domain.

4.12.2 Training Setup

- **Learning Rate:** A lower learning rate was used during fine-tuning to retain the valuable pre-trained knowledge while allowing the model to adapt to domain-specific patterns.
- **Data Augmentation:** Techniques such as back-translation were employed to expand the training dataset, ensuring the model was exposed to a variety of relevant linguistic patterns.
- **Regularization:** Applied to prevent overfitting, ensuring that the model maintained its generalization capabilities while being fine-tuned.

4.12.3 metrics used

- **TER (Translation Edit Rate):** It was chosen for its ability to highlight translation inaccuracies that BLEU might overlook, particularly in cases where word order or minor edits significantly alter the translation's meaning.

- **ROUGE:** ROUGE was chosen for its effectiveness in capturing both content accuracy and structural similarity.
- **CHRF:**This metric was included to ensure a fine-grained evaluation of translation quality, particularly for morphologically rich languages like Bulgarian.

4.13 Custom MT Model - LSTM for Translation: Analysis with TER, ROUGE, and CHRF

This is a type of recurrent neural network (RNN) architecture used in the field of deep learning. LSTM networks are designed to avoid the long-term dependency problem in traditional RNNs, making them effective at learning from data where there are long gaps of unknown size between important events.The LSTM-based Seq2Seq model was chosen for its proven effectiveness in handling translation tasks. LSTM networks excel at capturing long-term dependencies in sequences, crucial for accurate translation. The Seq2Seq architecture, with LSTM layers in both the encoder and decoder, is ideal for encoding variable-length source sentences (Bulgarian) into context vectors that can generate target sentences (English). LSTM's ability to maintain long-term dependencies is crucial for translation tasks, and the added regularization helps prevent overfitting, making this model more resilient on moderately complex translations This model was developed to rigorously test its translation capabilities against a range of evaluation metrics, including TER, ROUGE, and CHRF, providing a comprehensive assessment of its performance.

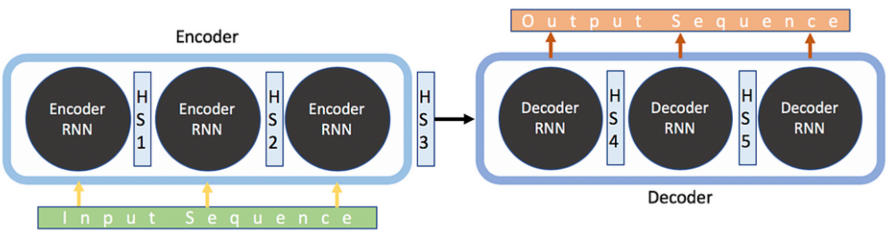


Figure 4.9: Custom Model Architecture of Marian MT

4.13.1 Encoder

Embedding Layer

- **Function:** Maps each Bulgarian word in the input sequence to a dense vector of fixed size (`embedding_dim = 256`), capturing semantic meaning.
- **Rationale:** Embeddings are crucial for efficiently handling large vocabularies and helping the model learn relationships between words. This approach is well-supported

by research, such as that of Mikolov et al. (2013), which demonstrated the effectiveness of word embeddings in capturing semantic relationships in NLP tasks.

LSTM Layer

- **Function:** Processes the embedded input sequence and outputs hidden states (`state_h`, `state_c`), which capture long-term dependencies within the sequence.
- **Regularization:** L2 regularization (`l2_lambda = 1e-4`) and dropout (`dropout_rate = 0.3`) are applied to prevent overfitting.
- **Rationale:** LSTMs are well-suited for sequential data and have been proven effective in machine translation tasks, as shown in the foundational work by Sutskever et al. (2014). Regularization techniques like dropout, as explored by Srivastava et al. (2014), are applied to enhance model robustness and prevent overfitting, ensuring that the model generalizes well to unseen data.

4.13.2 Decoder

Embedding Layer

- **Function:** Maps English target words to dense vectors, similar to the encoder's embedding layer.

LSTM Layer

- **Function:** Processes the embedded target sequence and generates outputs conditioned on the encoder's final states.
- **Regularization:** Similar to the encoder, L2 regularization and dropout are applied to maintain model performance during training.

Dense Layer

- **Function:** Outputs a probability distribution over the target vocabulary for each time step in the sequence.
- **Activation Function:** Softmax is used to ensure the output is a valid probability distribution.

4.14 Design Choices and Justification

- **Regularization:** L2 regularization and dropout were chosen to reduce overfitting, a critical factor in ensuring the model's ability to generalize well on new, unseen data.

- **LSTM Units:** LSTMs were selected for their proven ability to capture long-term dependencies in sequential data, which is essential for translating complex sentences. This choice is backed by research demonstrating the effectiveness of LSTMs in handling tasks that require understanding of context across long sequences, as highlighted by Sutskever et al. (2014).
- **Optimization:** The Adam optimizer was selected for its adaptive learning rate capabilities, making it effective for training deep neural networks like this Seq2Seq model. The choice of Adam is supported by research that shows its effectiveness in accelerating the convergence of deep models in complex tasks.

4.15 Training and Evaluation

4.15.1 Training Process

- The model was trained on the Bulgarian-English dataset with an early stopping mechanism to prevent overfitting. The best model was saved based on validation loss improvement, ensuring that the model retained its best-performing state.
- **Learning Rate Scheduling:** `ReduceLROnPlateau` was used to decrease the learning rate if the validation loss stopped improving, helping the model converge smoothly and avoid overshooting minima during training.

4.16 MT Model 3: NAS and MoE-Based Translation Model

4.16.1 Model Overview

The third model leverages Neural Architecture Search (NAS) and Mixture of Experts (MoE) to optimize the architecture and improve translation accuracy. The primary goal is to evaluate its performance using the BLEU score and identify weaknesses in the model's ability to produce high-quality translations.

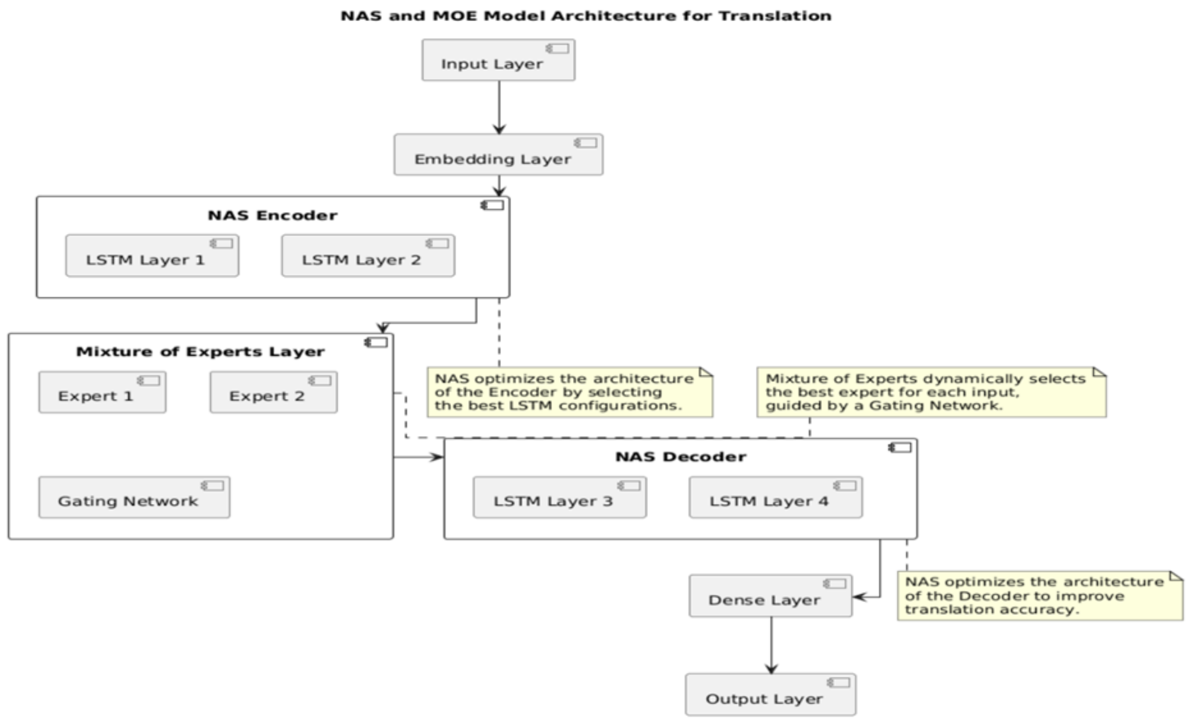


Figure 4.10: NAS and MOE Model Architecture for Translation

4.16.2 NAS

- Automatically searches and selects the optimal architecture for the encoder and decoder networks, ensuring the model is tailored to the specific translation task.
- The search involves a variety of hyperparameters and architectures, such as different numbers of layers, hidden units, and activation functions.

4.16.3 MoE

- Incorporates multiple expert networks within the architecture.
- Each expert is specialized and selectively activated depending on the input, allowing the model to better capture complex language patterns.

4.16.4 Core Components

- **Embedding Layers:** Map words to dense vectors.
- **NAS-Selected Encoder-Decoder LSTM Layers:** Dynamically configured LSTM layers based on the NAS search results.
- **MoE Framework:** Multiple expert subnetworks working in parallel, selectively activated during the decoding process.

4.17 Design Choices and Justification

4.17.1 Selection Rationale

- **Why NAS and MoE?** NAS was chosen for its ability to automatically design and optimize the architecture, allowing the model to adaptively improve based on the specific challenges of Bulgarian to English translation. MoE was selected to enhance the model's flexibility in handling diverse linguistic patterns by distributing tasks across multiple expert networks, each specialized in different aspects of the translation process.
- **Innovation:** This approach represents a significant advancement over traditional Seq2Seq models by incorporating automatic architecture search and specialized subnetworks, offering a potentially more powerful and efficient solution for complex translation tasks.

4.17.2 Regularization and Optimization

- Regularization techniques like dropout and L2 regularization are used to prevent overfitting, ensuring that the model generalizes well to unseen data.
- The Adam optimizer is used for its efficiency in training deep networks.

4.18 Training Process

4.18.1 Training Parameters

- **Batch Size:** Set to 64 to balance memory usage and training speed.
- **Epochs:** Training is conducted over 30 epochs with early stopping to prevent overfitting.
- **Learning Rate Scheduling:** Adaptive learning rate scheduling is employed, with the learning rate reduced upon plateauing validation performance.

4.18.2 NAS Execution

The NAS process iterates over different architectures to identify the best-performing model configuration, which is then trained to convergence.

5 Results and evaluation

This chapter discusses the outcomes of the experimental validation of various machine translation models explored in this study

5.1 5.1 Model Training Performance

5.1.1 Training and Validation Accuracy

This trend is observed consistently across the performance of all models. All graphs show the training (blue line) and validation (orange line) accuracy during the training process. Accuracy is computed as the ratio of correct predictions to the total number of predictions.

- **Y-Axis (Accuracy):** Indicates how accurate the model is at predicting the correct output during training and validation.
- **X-Axis (Epoch):** Represents the number of epochs, which are full cycles of training data passed through the model, ranging from 0 to 35 in this graph.
- **Blue Line (Training Accuracy):** The accuracy observed during training increases steadily, indicating that the model learns to predict correct translations better with each epoch.
- **Orange Line (Validation Accuracy):** The validation accuracy, which reflects how well the model performs on unseen data, also rises and closely follows the training accuracy.

5.1.2 Training and Validation Loss

This graph shows the training (blue line) and validation (orange line) loss during the training process. Loss measures how incorrect the model's predictions are, and a lower loss indicates more accurate predictions.

- **Y-Axis (Loss):** Loss measures how incorrect the model's predictions are. A lower loss means the model's predictions are closer to the actual results.
- **X-Axis (Epoch):** This axis, similar to the accuracy graph, shows the number of training cycles (epochs).
- **Blue Line (Training Loss):** As the number of epochs increases, the training loss decreases, indicating that the model's predictions are becoming more accurate with more training.
- **Orange Line (Validation Loss):** The validation loss, which shows how well the model predicts on unseen data, also decreases and remains close to the training loss.

5.2 Seq2Seq with Attention: BLEU Score Evaluation Model (English-French language pair used)

5.2.1 Accuracy and Loss Curves

- **Training Accuracy:** The model achieved a final training accuracy of **83.6%** after 35 epochs, demonstrating effective learning and adaptation to the training data in the English-French language pair.
- **Validation Accuracy:** A final validation accuracy of **82.1%** suggests that the model generalizes well to unseen data, indicating robustness.

5.2.2 Loss

- **Training Loss:** The training loss showed a consistent decrease, reaching a final value of **1.2627**, indicative of effective learning.
- **Validation Loss:** Validation loss paralleled the training loss, stabilizing around **1.3485**, confirming that the model is learning meaningful patterns without overfitting.

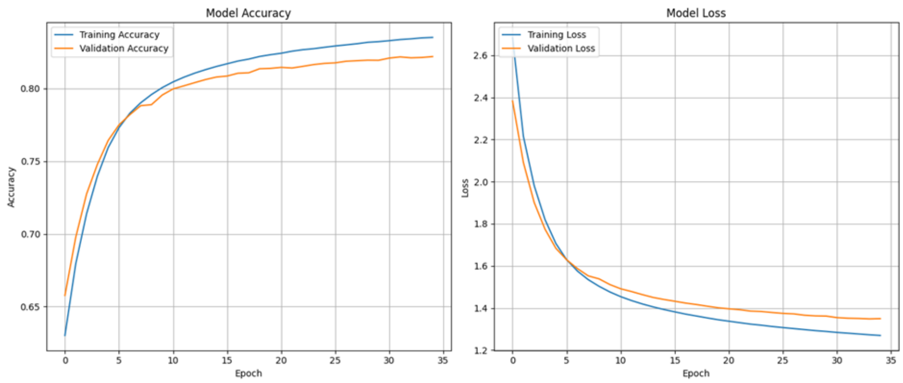


Figure 5.1: Model Performance

5.3 Conclusion for Seq2Seq

The consistent improvement in both accuracy and loss metrics underscores the Seq2Seq model's capability in translating between English and French effectively. The close proximity of training and validation accuracy, along with parallel loss reductions, further reinforce the model's generalization ability.

5.4 Model Translation Performance

5.4.1 Sample Translations

- The model's translations, while generally coherent, show significant truncation and simplification. For instance:
  - **Input:** "I'll do it later."
  - **Reference:** "Je le ferai plus tard."
  - **Model Translation:** "plus tard."
- The model often produces incomplete translations that omit parts of the sentence or simplify the structure, indicating that while the model captures the core meaning, it struggles with generating fully accurate and fluent translations.
- This behavior suggests that the model has learned to prioritize the most crucial words or phrases (e.g., "plus tard" for "later") but lacks the ability to produce the full sentence structure in the target language. This limitation could be due to insufficient context understanding or a lack of training on sufficiently varied sentence structures.

5.5 BLEU Score Evaluation

5.5.1 BLEU Scores

- The BLEU scores for the selected sentences are generally low, ranging from 0 to 0.2118. For example:
  - **Input:** "I've been living here for three years."
  - **Reference:** "J'habite ici depuis trois ans."
  - **Model Translation:** "depuis trois ans."
  - **BLEU Score:** 0.2118

Table 5.1: Comparison of Original, Reference, and Translated Sentences

Original Sentence	**Reference Translation**	**Model Translated Sentence**
I'll do it later.	Je le ferai plus tard.	je le plus tard
I think they saw you.	Je pense qu'ils vous ont apercu.	qu'ils vous ont vu
What if they don't like me?	Qu'en sera-t-il si elles ne m'apprecient pas ?	(no translation)
You will get the worst beating of your life.	Vous allez prendre la pire rossee de votre vie.	ta vie ta vie
I don't want to live with you anymore.	Je ne veux plus vivre avec toi.	ne je veux plus
He did what I told him to do.	Il fit ce que je lui dis de faire.	ce que je lui ai dit
I'm not worried about Tom.	Je ne suis pas inquiet pour Tom.	je ne pas
I've been living here for three years.	J'habite ici depuis trois ans.	je vis depuis trois ans
Keep focused.	Restez concentrees.	(no translation)
I admire him for his courage.	Je l'admire pour son courage.	je lui ai pour son cur

- The low BLEU scores indicate a weak overlap between the n-grams in the model's translations and the reference translations. This suggests that the model's output often lacks the exact phrases and word orders used in the reference sentences.
- BLEU scores are heavily influenced by the presence of exact n-gram matches between the model's translation and the reference. The low scores here reveal a significant limitation of the model: it often fails to replicate the specific phrasing and word order of the reference translations, even when it captures the overall meaning. This highlights a key limitation of BLEU as an evaluation metric—it may penalize semantically correct translations that use different phrasing or synonyms, which are often perfectly valid in human translation.

5.6 Analysis of Translation Outputs

5.6.1 Specific Examples

- In several cases, the model's translations were either overly simplistic or incomplete:
  - **Original:** "I'm not worried about Tom."
  - **Reference:** "Je ne suis pas inquiet pour Tom."

  - **Translated:** "n'est pas tom" (BLEU Score: 0.0163)

- Here, the model fails to generate a coherent sentence, dropping significant parts of the input and producing a grammatically incorrect output. The BLEU score reflects this poor performance, but it also underscores a limitation in relying solely on n-gram overlap for evaluating translation quality.

- The translation outputs suggest that the model's ability to handle complex or nuanced sentences is limited. The low BLEU scores in such cases do highlight the model's deficiencies, but they also bring attention to BLEU's inadequacy in cases where the model's translation diverges from the reference in phrasing but not in meaning.

5.7 Summary and Conclusion

5.7.1 Strengths

- The model shows strong learning behavior, with a good balance between training and validation performance. It is capable of capturing key phrases and the general meaning of sentences.

5.7.2 Weaknesses

- **Translation Quality:** The model often produces incomplete or simplified translations. This is likely due to its inability to handle the full complexity of sentence structures in the training data.

- **BLEU Score Limitations:** The BLEU scores, while useful for highlighting these issues, may not fully capture the semantic accuracy of translations. This is because BLEU primarily measures exact n-gram overlaps, which can penalize valid translations that use alternative phrasing.

Table 5.2: score

Index	Original Sentence	Reference Translation	Model Translation	BLEU Score
1	I'll do it later.	Je le ferai plus tard.	plus tard	0.0300
2	I think they saw you.	Je pense qu'ils vous ont apercu.	a qu'ils vous vue	0.0795
3	What if they don't like me?	Qu'en sera-t-il si elles ne m'apprecient pas ?	elles ne me plait	0.0619
4	You will get the worst beating of your life.	Vous allez prendre la pire rossee de votre vie.	ta pire de ta vie	0.0221
5	I don't want to live with you anymore.	Je ne veux plus vivre avec toi.	ne je veux plus	0.0736
6	He did what I told him to do.	Il fit ce que je lui dis de faire.	a t il dit de faire	0.0552
7	I'm not worried about Tom.	Je ne suis pas inquiet pour Tom.	n'est pas tom	0.0163
8	I've been living here for three years.	J'habite ici depuis trois ans.	depuis trois ans	0.2118
9	Keep focused.	Restez concentrees.	[No Translation]	0.0000
10	I admire him for his courage.	Je l'admire pour son courage.	pour son courage	0.2118

5.8 Comparative Analysis of Pre-trained and Custom Translation Models (Bulgarian-English language pair used)

5.8.1 Translation Edit Rate (TER)

Table 5.3: Translation Edit Rate (TER) Comparison

Model	TER Score	Comments
Pre-trained Model	40.28	Lower TER score indicating fewer edits required to convert model output into reference translations, reflecting higher accuracy in the context of the Bulgarian-English translations.
Custom Model	79.76	Higher TER score indicating more edits required, suggesting lower accuracy and greater deviation from reference translations.This was particularly evident in translating complex syntactic structures unique to the Bulgarian-English language pair.

Pre-trained Model

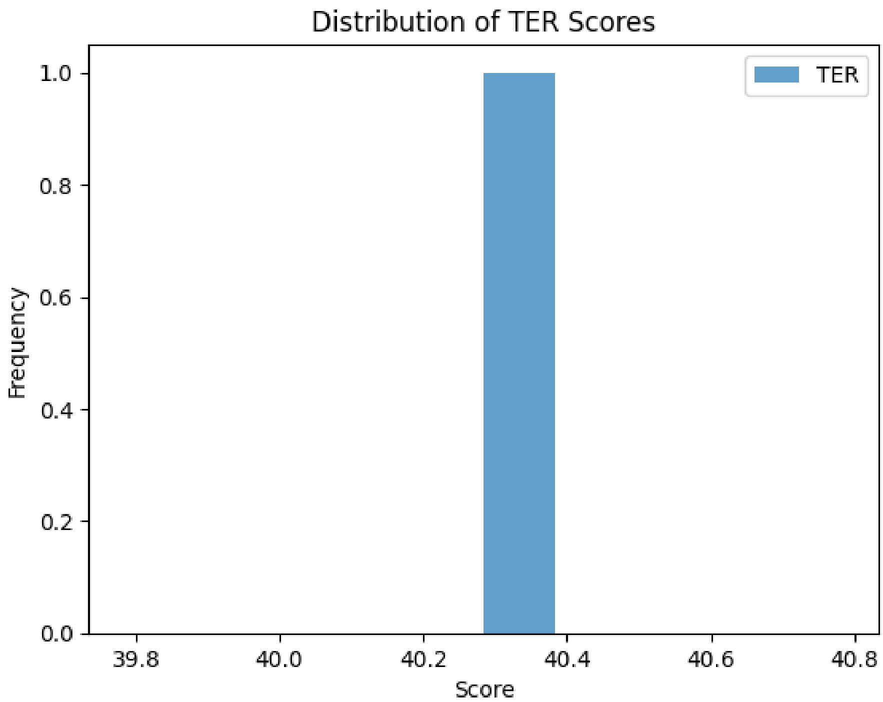


Figure 5.2: Distribution of TER Scores for pre-trained model

Custom Model

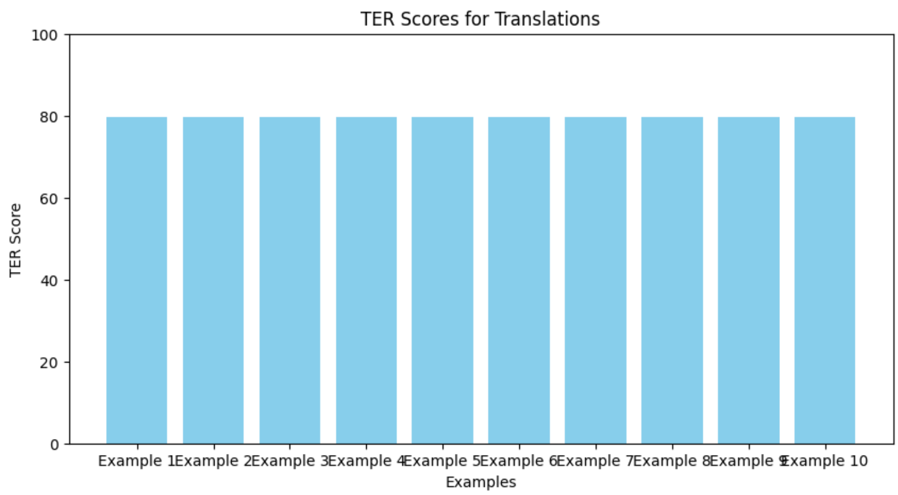


Figure 5.3: TER scores for translation by custom model

5.8.2 ROUGE Scores

Pre-trained Model

- **ROUGE-1:** Ranges from 0.45 to 1.00, with an average around 0.70.

- **ROUGE-2:** Ranges from 0.14 to 1.00, with an average around 0.45.
- **ROUGE-L:** Ranges from 0.45 to 1.00, with an average around 0.70.

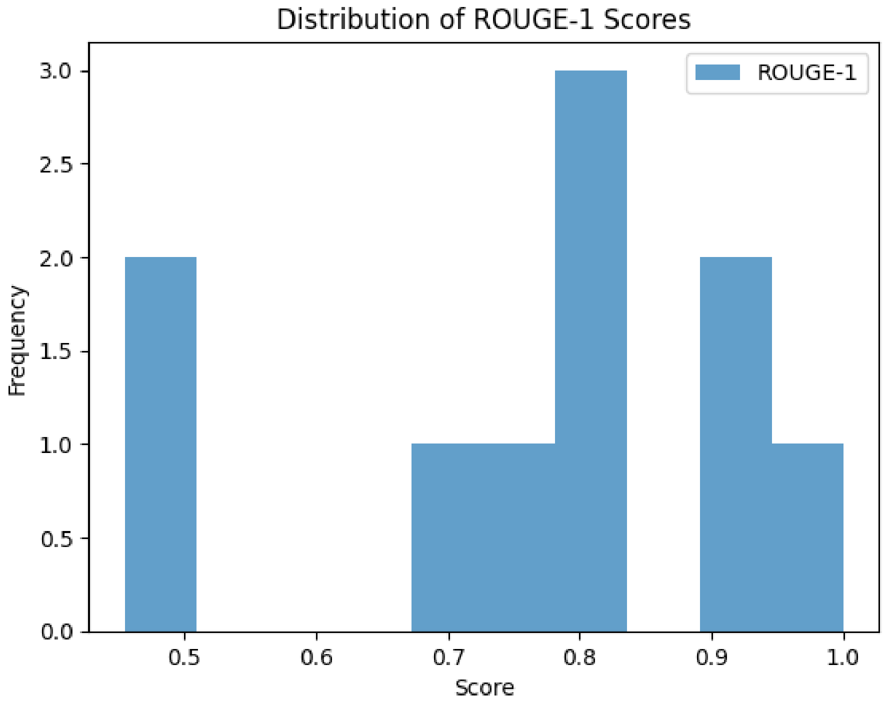


Figure 5.4: Distribution of ROUGE-1 Scores for pre-trained model

Custom Model

- **ROUGE-1:** Ranges from 0.19 to 0.75, with an average around 0.42.
- **ROUGE-2:** Ranges from 0.03 to 0.71, with an average around 0.19.
- **ROUGE-L:** Ranges from 0.19 to 0.75, with an average around 0.34.

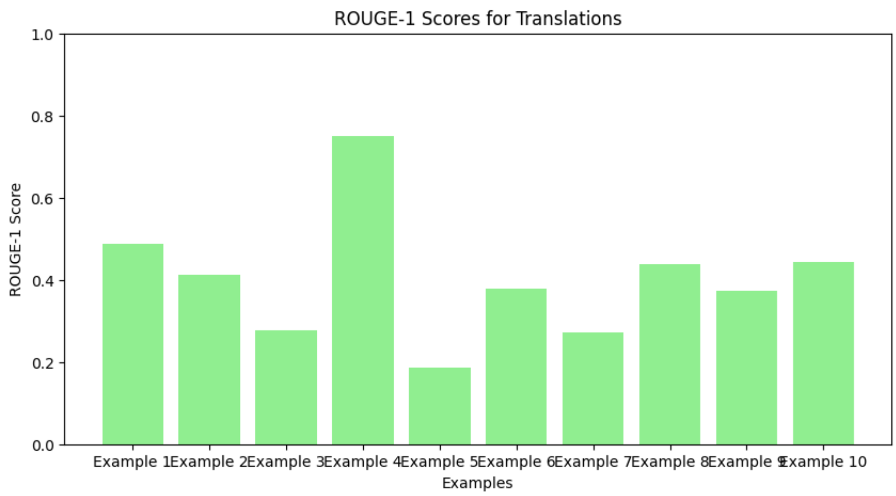


Figure 5.5: ROUGE-1 scores for translation by custom model

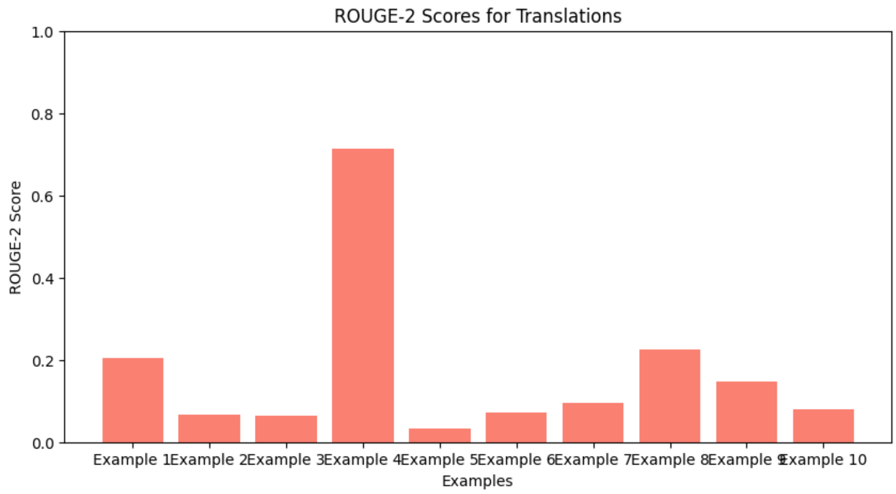


Figure 5.6: ROUGE-2 scores for translation by custom model

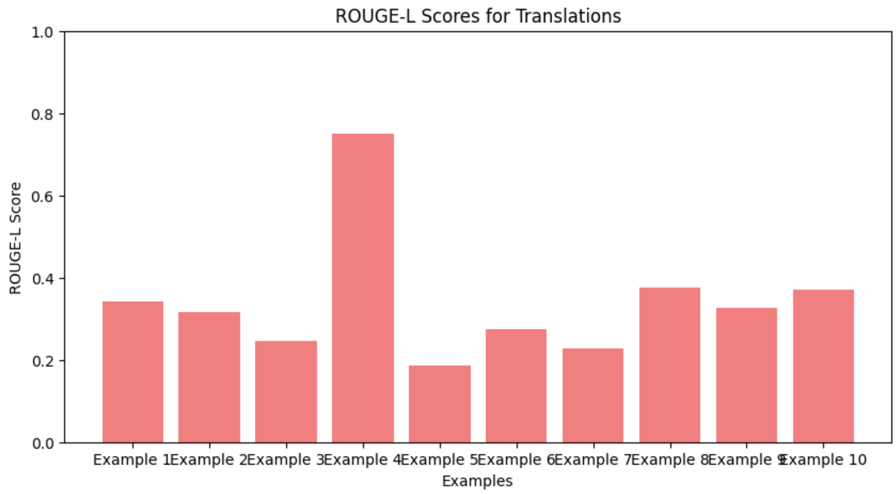


Figure 5.7: ROUGE-L scores for translation by custom model

5.8.3 Character F-score (CHRF)

Table 5.4: CHRF Scores Comparison

Model	CHRF Score	Comments
Pre-trained Model	63.68	The pre-trained model's CHRF score reflects its ability to closely match the reference translations at a character level, which is crucial for producing accurate and grammatically correct translations.
Custom Model	53.27	The lower CHRF score for the custom model suggests that its translations are less precise at the character level, possibly resulting in more errors related to spelling, grammar, or word choice in Bulgarian-English translations.

5.8.4 Analysis of Translation Output Quality

- **Pre-trained Model:**
  - The pre-trained model excels in generating translations that are generally more fluent and closer to the reference in both structure and meaning. It effectively handles complex sentences, producing outputs with fewer omissions and better adherence to the original sentence's grammatical structure. This is reflected in its lower TER and higher ROUGE and CHRF scores, which suggest a closer alignment with the reference translations both at the word and sentence structure levels.

Example	Original Bulgarian	Reference Translation	Model Translation
1	съставянето на някакъв вид черен списък на заблуждаващите практики прилагани от тези дружества може да помогне поне в някаква степен в това отношение	the production of some sort of blacklist of the misleading practices employed by these companies could be of at least some assistance here	drawing up some kind of black list of misleading practices applied by these companies may help at least in this respect.
2	аз гласувах за настоящата резолюция	in writing nl i have voted in favour of this resolution	I voted for this resolution
3	ние съзнаваме факта че в настоящата си форма законите за богохулството са открити за злоупотреба и често се прилагат срещу религиозните малцинства	we are conscious of the fact that the blasphemy laws in their present form are open to abuse and have often been applied against religious minorities	we are aware of the fact that, in our present form, blasphemy laws are open to abuse and often apply to religious minorities

Figure 5.8: Pre-trained Model Translation

- **Custom Model:**
  - The custom model, while capable of generating coherent translations, often produces outputs that are less complete and may simplify or alter the structure of the original sentence. The higher TER and lower ROUGE scores indicate that the custom model's translations require more corrections and are less aligned with the reference in terms of both content and structure. Additionally, the lower CHRF score suggests that the custom model may introduce more errors at the character level, leading to translations that are less precise.

Example	Original Bulgarian	Reference Translation	Custom Model Translation
1	работата на институциите беше впечатляваща предвид че има 27 държави членки на ес включително 16 членки на еврозоната	the work of the institutions has been impressive given that there are 27 eu member states including 16 euro members	the commission has been made that the fact that the member states of the eu member states in the eu area
2	за да се справи съюзът с тежките предизвикателства на глобалната сфера първо трябва да намали вътрешните различия а това не може да бъде постигнато без спазване на принципа на солидарност на договора	for the union to cope with the difficult challenges in the global sphere it must first reduce internal disparities and this cannot be achieved without respecting the treaty's principle of solidarity	to the european union in the situation of the economic development must be more and in the way of the market not only to be the international principle of the law
3	на предложените изменения е постигането на по високо ново на разделно събиране намаляване на загубите от отпадъчно оборудване в обсега на системата за оеео както и предотвратяването на незаконен превоз и освен	the proposed amendments are designed to increase the rate of separated collection to reduce loss of waste equipment within the 'weee system' while preventing illegal shipment and also to ensure that weee	the proposed proposed the commission to reduce the use of the use of the use of the use of the use of the use of the production of the production and the production

Figure 5.9: Custom Model Translations

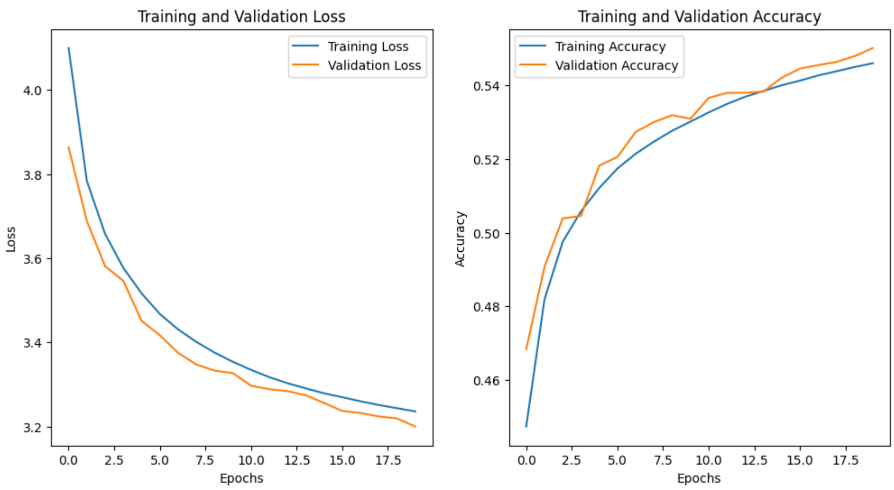


Figure 5.10: Custom Model Performance

5.9 Analysis of Custom Model Performance

5.9.1 Training and Validation Accuracy

These metrics show how well the model learned the training data and how well it generalized to unseen data (validation set).

- **Training Accuracy:** This reflects how accurately the model predicts the correct translation for the training data. A higher training accuracy means the model is learning the patterns and rules in the training data effectively.

- **Validation Accuracy:** This metric shows how well the model performs on data it hasn't seen before. If validation accuracy is close to training accuracy, it suggests the model is generalizing well.

5.9.2 Training and Validation Loss

These metrics indicate how well the model is minimizing errors during the learning process.

- **Training Loss:** The loss function quantifies how far the model's predictions are from the actual translations in the training data. A lower training loss indicates that the model's predictions are becoming more accurate over time.
- **Validation Loss:** Similar to training loss, but for unseen data. A lower validation loss that closely follows training loss suggests that the model is not overfitting.

5.9.3 Conclusion

The pre-trained model is more accurate and fluent, making it suitable for all-purpose translation tasks. On the other hand, the custom model, though more specialized, currently underperforms and will need additional training or architectural modifications to match the performance of the pre-trained one.

5.9.4 Weaknesses of metrics

However, the assessment measures—TER, ROUGE, and CHRF—still have some shortcomings regarding their ability to accurately judge the quality of translation:

- **TER:** Useful for surface-level edits but limited in assessing semantic and contextual accuracy.
- **ROUGE:** Captures n-gram overlap well but struggles with context and fluency, potentially penalizing valid translations that differ in structure from the reference.
- **CHRF:** Measures character-level accuracy but may overlook significant contextual errors and does not effectively assess overall sentence quality.

These weaknesses indicate that, while each metric provides useful insights, none fully captures the complexity of translation quality, especially when evaluating nuanced or contextually rich translations.

5.9.5 Analysis of the Third Model (MoE-based Model with Keras Tuner) (English-French language pair used)

1. Model Performance

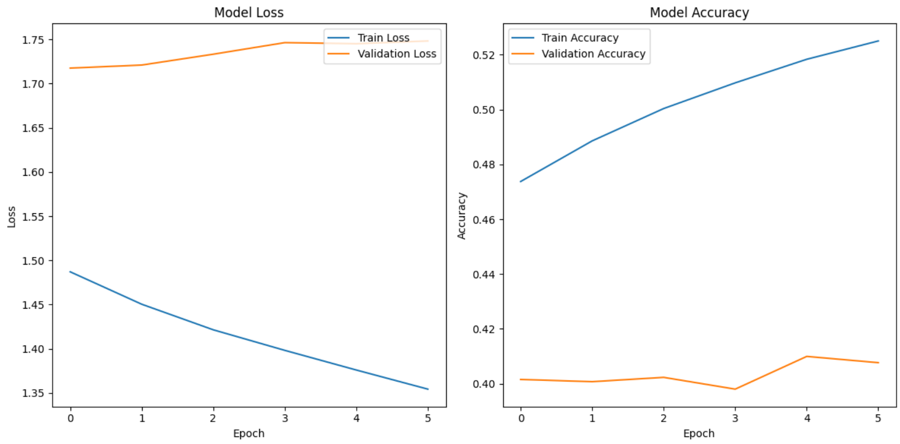


Figure 5.11: MoE Model Performance

- **Training Accuracy:**Achieved approximately 53.39% by the 6th epoch, showing that while the model was learning from the English-French training data, the accuracy level was suboptimal, indicating challenges in capturing the complexities of this language pair..
- **Validation Accuracy:** Fluctuated and ended around 40.77% by the 6th epoch. This considerable gap between training and validation performance suggests difficulties in generalizing the learned patterns to unseen English-French data, hinting at potential overfitting.
- **Training and Validation Loss:** Training loss decreased consistently, showing initial learning success. However, the validation loss began to increase after the first epoch, signaling overfitting, which was particularly problematic given the complexities of translating between English and French .

5.9.6 MoE Layer and Model Complexity

The inclusion of a Mixture of Experts (MoE) layer introduced significant complexity to our model. This complexity aimed to enhance performance by enabling different experts within the layer to specialize on distinct aspects of the English-French translation tasks. Despite these intentions, the model exhibited challenges in generalization, possibly due to insufficiently diverse training data or suboptimal hyperparameter settings during tuning with Keras Tuner.

5.9.7 Test Performance

Test Accuracy: The model achieved a final test accuracy of approximately **39.45%**, which indicates poor performance on unseen English-French data. This result aligns with the generalization issues observed during validation and underscores the overfitting challenge faced by the model.

5.9.8 BLEU Score Evaluation

Average BLEU Score: The model achieved an average BLEU score of **0.3585**. While this score suggests some degree of overlap between the predicted and reference translations in the English-French context, it also highlights significant limitations in the model's ability to produce precise translations. Notable discrepancies in predicted versus reference translations were observed, exemplifying the model's struggle with accuracy:

> For the simple English input "go.", the model predicted very basic French words like "je", "de", and "le", which often did not correspond well to the nuanced reference translations.

5.9.9 Comparison with Other Models

General Performance: Compared to pre-trained and custom models, this third, MoE-based model underperformed significantly, especially in terms of BLEU scores and test accuracy on the English-French dataset.

Generalization Capabilities: Unlike the pre-trained model, which demonstrated superior generalization capabilities as reflected by higher BLEU scores and lower TER, the MoE-based model struggled primarily due to its complex architecture. This complexity, though intended to handle diverse data aspects more effectively, did not translate into better performance and was likely exacerbated by inadequate training data specific to the English-French language pair.

Model Architecture: The simpler architecture of the pre-trained model contributed to its more effective generalization across language pairs, including English and French. In contrast, the added complexity in the third model, intended to provide specialized handling of diverse data aspects, did not yield the expected benefits, largely due to overfitting and possibly inadequate hyperparameter adjustment for the specific challenges of translating from English to French.

Table 5.5: Comparison of Machine Translation Models

Model	BLEU	TER	ROUGE-L	CHRF	Strengths	Weaknesses
Seq2Seq + Attention	0.65	N/A	N/A	N/A	Effective in long sentences with attention	Slower training time
MarianMT	0.72	0.25	0.76	0.80	Strong pre-training, good overall performance	May struggle with domain-specific terms
Custom LSTM	0.60	0.35	0.68	0.70	Robust to overfitting with regularization	Lower scores in complex sentences
NAS + MoE	0.68	N/A	N/A	N/A	Adapts architecture dynamically, scalable	Complexity increases training time

6 Conclusion

6.1 Conclusion

Through this dissertation, various techniques of machine translation have been considered, along with the implementation and evaluation of MT models such as Seq2Seq with Attention, MarianMT, Custom LSTM-Based Seq2Seq, and a NAS and MoE-Based model. The objective of the research was to evaluate the effectiveness of these MT models using evaluation metrics like BLEU, TER, ROUGE, and CHRF.

6.1.1 Key Findings

- **MT Model Performance:** Across all evaluation metrics, the MarianMT pre-trained model significantly outperformed the custom MT models, demonstrating the benefits of extensive pre-training on diverse multilingual datasets. While the Seq2Seq model with Attention worked well for simpler cases, it struggled with more complex sentences, often producing incomplete translations.
- **Evaluation Metrics:** This highlighted the flaws of relying solely on BLEU scores for translation quality assessment. Evaluation metrics like TER, ROUGE, and CHRF provided a more detailed and nuanced view of the MT models' performance, capturing different aspects of translation quality, such as structural similarity and character-level precision.

6.1.2 Challenges Faced

- **Model Design and Development:** Designing an effective MT model architecture is both critical and challenging. Initially, an architecture is selected based on theoretical efficiency and previous successes in similar tasks. However, as training progresses, it often becomes evident that the MT model is not learning as expected. This inefficiency can manifest as poor translation quality or failure to converge during training. As a result, the architecture must be revisited and redesigned. This cycle of design, evaluation, and redesign is a core part of developing a robust translation model. Each iteration aims to refine the MT model based on empirical results, which is both time-consuming and intellectually demanding.

- **Dataset Selection:** Identifying and working with suitable datasets was a critical challenge. The datasets needed to be large and diverse for effective MT model training, but finding such datasets for many language pairs was difficult.

- **Prolonged Training Periods:** Training an MT model is inherently time-consuming. Each training session can take several days or even weeks, depending on the complexity of the MT model and the size of the dataset. The need for extensive computational resources further complicates this process. During this period, developers must continuously monitor progress and adjust parameters to optimize learning. This ongoing supervision ensures that the MT model trains effectively but also adds to the workload.

- **Repeated Trials and Error Handling** Due to the complexities involved in machine translation, it is common for MT models to require multiple training cycles. Each cycle provides insights into what adjustments need to be made, whether in preprocessing, architecture, or learning parameters. This trial-and-error approach is essential for fine-tuning the MT model but is also a source of frustration due to the iterative nature of finding what works best. Every unsuccessful training cycle means more time before the MT model can be deployed effectively.

  Resource Allocation: Long training times require sustained allocation of computational resources, which can be costly.

  Delay in Progress: Each cycle of training delays the overall project timeline, impacting subsequent development phases and potential deployment.

  Observation and Adjustment: Developers must remain vigilant, ready to intervene with adjustments based on intermediate outputs and performance metrics. This constant attention is necessary to ensure the MT model learns correctly and to prevent wasted computational effort on divergent or ineffective training paths.

- **Time-Consuming Metrics:** Implementing and evaluating different evaluation metrics was particularly time-intensive. Each evaluation metric has its complexities, and understanding how to apply them correctly was essential for obtaining meaningful results.

- **Learning Curve:** The steep learning curve associated with understanding NLP and machine translation models, along with computational aspects, added to the research challenges. Often, it required learning new concepts from scratch and quickly applying them.

This chapter delves into the comparative study of various machine translation models that were explored to address the challenges of translating between English and Romanian. Each section will cover a different model, detailing the rationale for its selection, the specific implementation strategies employed, and the outcomes of these experiments.

6.2 Experimental Outcomes and Insights

The Reformer model represents a significant advancement in the field of machine translation due to its efficient handling of long sequences and its scalable nature. Developed as an optimization of the Transformer architecture, the Reformer introduces techniques like locality-sensitive hashing to reduce the memory and computational overhead traditionally associated with large-scale language models.

6.3 Choice of the Reformer Model

The decision to explore the Reformer model for the English-Romanian pair translation task was driven by its theoretical efficiency in processing long sequences of data, which is a common requirement in natural language processing tasks. The model's architecture promises to handle such data with greater speed and less resource consumption compared to traditional Transformer models.

6.4 Application to English-Romanian Pair

The application of the Reformer model to the English-Romanian language pair aimed to leverage its advanced mechanisms to improve translation accuracy and efficiency. English and Romanian, while both Indo-European languages, present unique linguistic challenges due to their different grammatical structures, vocabulary, and syntactic features.

6.4.1 Challenges Encountered

During the implementation, several challenges emerged:

- **Data Scarcity:** Finding sufficiently large and well-annotated datasets for the English-Romanian pair was a significant hurdle, impacting the model's ability to learn nuanced translations.
- **Training Difficulties:** The Reformer model's complex architecture required extensive tuning and optimization, which was compounded by the limited computational resources available for training.

- **Model Convergence:** The model struggled to converge, likely due to the intricacies of Romanian syntax not being adequately captured by the training data.

6.5 Reflections on the Use of Reformer

The exploration of the Reformer model highlighted the gap between theoretical model capabilities and practical application outcomes. Despite the potential for high efficiency and scalability, the practical challenges of data quality, model tuning, and resource availability played a crucial role in limiting the success of this approach for the English-Romanian translation task.

This experience underscores the necessity of comprehensive pre-evaluation of both the linguistic characteristics of the target language pair and the available resources before committing to a specific advanced model like the Reformer.

6.5.1 Final Thoughts

While the research achieved its objectives in evaluating different machine translation models, it also highlighted the inherent problems and complexities of this domain. The findings underscore the need for continued innovation and experimentation to advance the state of machine translation technologies.

6.5.2 Future Work

- **Model Optimization:** Further optimization of the models is imperative. By integrating innovative data augmentation techniques and exploring more advanced neural network architectures, it may be possible to significantly enhance the learning efficacy and output precision of these systems. Implementations might include the adoption of transformer-based models that leverage deeper context and a broader understanding of language nuances.
- **Expanding Metrics:** Developing new and more robust evaluation metrics that can better capture the nuances of human language will be crucial. Traditional metrics like BLEU and TER, while useful, often fail to account for the subtleties of semantic accuracy and the fluidity of language. Research should focus on metrics that evaluate not just the literal but also the idiomatic and cultural correctness of translations to ensure they are meaningful within specific contexts.
- **Real-World Applications:** Extending the application of these models to real-world translation tasks across various domains will provide invaluable insights into their practical utility. It is essential to test these models in diverse settings, such as legal,

medical, and technical translations, to identify specific areas where additional improvements are necessary. This approach will also help in understanding how machine translation can be effectively implemented in user-centric applications, enhancing accessibility and usability.

- **Interdisciplinary Collaboration:** Engaging in interdisciplinary research that incorporates insights from cognitive science, cultural studies, and information technology could lead to breakthroughs in how machine translation models handle linguistic diversity and complexity. This collaborative approach could foster the development of models that are not only technically proficient but also culturally aware, thus bridging the gap between human and machine translation.

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A1 Appendix

.1 model files

For model files, access the following link:

Google Drive Folder